

On-device cough detection and respiratory disease classification enhanced by generative data augmentation

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HIGHLIGHTS

- On-device smartphone framework for privacy-preserving respiratory cough monitoring.
- Lightweight 1D-CNN (90 kB) detects cough events in real time at 90.0% ratio.
- Parallel SVMs with Bayesian fusion classify asthma, COVID-19, and healthy coughs.
- Dual-domain VQ-VAE augmentation lifts disease attribution macro-F1 by 6 pp to 81.1%.
- Abstain rule ($\tau = 0.80$) yields 91.7% macro-F1 at 74.2% coverage on covered set.

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ABSTRACT

Background: Cough sounds are accessible, non-invasive biomarkers for respiratory disease assessment and can be captured using consumer-grade smartphones. Existing approaches typically focus solely on cough detection or rely on server-based deep learning for disease classification, which limits deployability and raises privacy concerns. Small, imbalanced cough datasets further hinder model generalization.

Objective: To develop a multilayer, smartphone-compatible AI framework for automated cough detection and respiratory disease classification, and to propose a pioneering generative augmentation strategy utilizing a suite of five Variational Autoencoder (VAE) variants and a probabilistic cough-level fusion mechanism to improve disease classification under severe data scarcity and the limitations of conventional audio augmentation techniques.

Methods: The proposed framework consists of three AI modules: (1) A *Cough Detection Module (CDM)* that performs real-time cough event detection and segmentation from continuous audio using lightweight models optimized for on-device execution. (2) A *Disease Analysis Module (DAM)* that classifies cough events into asthma, COVID-19, or healthy classes using parallel Support Vector Machine classifiers and a probabilistic cough-level fusion strategy. (3) A *Generative Augmentation Module (GAM)* employing five distinct VAE architectures. This module uniquely operates across the time-frequency domain for latent feature optimization while reconstructing samples in the time-domain to ensure acoustic verifiability.

Results: The CDM provides reliable segmentation across heterogeneous recording conditions. The DAM achieves strong discriminability between asthma, COVID-19, and healthy coughs using compact cepstral and spectral features. The multi-variant GAM framework demonstrates superior efficacy in alleviating class imbalance specifically, the cross-domain (time-frequency to time-domain) reconstruction allows for the clinical verification of synthetic biomarkers. All system components operate in real time on commodity Android hardware.

Conclusion: This integrated framework addresses key limitations in dataset availability, model generalizability, and deployability, introducing an interpretable generative approach that demonstrates the feasibility of smartphone-based acoustic sensing as a scalable and privacy-preserving tool for respiratory health monitoring.

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1. Introduction

The analysis of respiratory sounds has gained substantial attention following the COVID-19 pandemic due to its capacity for non-invasive, cost-effective, and scalable disease screening [1]. Among these, cough sounds are of particular clinical and diagnostic relevance; they represent a prominent and easily captured acoustic biomarker associated with a wide spectrum of respiratory conditions [2]. Traditional diagnostic modalities, such as electronic stethoscopes and spirometry, face limitations in accessibility, cost, and reliance on trained personnel [3]. These constraints have accelerated the adoption of Artificial Intelligence (AI) and audio signal processing as promising tools for automated respiratory diagnosis [1].

A natural extension of this trend is the use of smartphones as ubiquitous sensing platforms for passive respiratory monitoring. Modern consumer devices, equipped with high-quality microphones and edge-computing capabilities, offer an opportunity to deliver scalable respiratory assessment without specialized hardware. Barata et al. [4] validated a contact-free smartphone-based cough monitor in hospital wards, demonstrating that commodity devices can provide reliable monitoring in low- and middle-income countries. Similarly, the Hyfe Cough Tracker demonstrated that convolutional models can run unobtrusively in the background of a smartphone for extended periods, enabling passive surveillance at negligible deployment cost [5].

Smartphone-based health sensing extends beyond cough analysis. Valentine et al. [6] demonstrated that smartphone inertial sensors can accurately estimate respiratory rate, providing an accessible alternative to dedicated remote-monitoring devices. Faezipour et al. [7] further argued that mobile respiratory assessment enables convenient self-testing and early detection of breathing disorders. At a population level, Gabaldon et al. [8] proposed digital cough surveillance powered by smartphone AI models, highlighting its potential as a scalable indicator of disease incidence. Collectively, these studies underscore the feasibility and public-health value of leveraging mobile devices for continuous monitoring across clinical and home settings.

Despite these advances, several gaps remain. Many existing systems focus solely on detecting or counting cough events without providing disease-specific analysis. Furthermore, studies that attempt cough classification are typically evaluated in controlled, offline settings and are not designed for real-world deployment constraints. Additionally, publicly available cough datasets are often small and imbalanced, which limits model generalization. The use of principled generative augmentation remains largely unexplored in this domain, particularly regarding frameworks capable of bridging the gap between high-dimensional latent feature learning and physically verifiable acoustic signals.

This work is motivated by a specific intersection of four requirements that, to the best of our knowledge, no prior smartphone cough system has jointly satisfied: (i) true on-device execution with no server dependency to preserve user privacy; (ii) disease-specific classification targeting asthma and COVID-19; (iii) a novel generative augmentation stage to mitigate the scarcity of disease-specific corpora; and (iv) the demand for clinical interpretability in synthetic data generation. While prior works have addressed ward-level counting [4,5] or cloud-based classification [1], the joint delivery of these features—specifically the ability to generate acoustically verifiable synthetic biomarkers, and their delivery on commodity hardware—remains an open problem in the current clinical landscape.

This joint requirement shapes our architectural choices: the Cough Detection Module utilizes a compact 1D-CNN to meet the latency budget of continuous background operation; the Disease Analysis Module employs parallel feature-based Support Vector Machines fused by a probabilistic voting mechanism for interpretability and efficiency; and the Generative Augmentation Module (GAM) introduces a multi-variant VAE ensemble. The GAM is uniquely designed to perform manifold learning in the time-frequency domain while reconstructing final

samples in the time-domain, allowing for “human-in-the-loop” acoustic verification.

Contributions. To address these gaps, we propose a unified, smartphone-native framework for passive cough-based health monitoring. Our contributions are as follows:

1. We design and implement an end-to-end module for passive, continuous cough detection and segmentation directly on a smartphone using a lightweight neural network and an explainable time-domain segmenter.
2. We develop a disease analysis module that classifies detected cough sounds into asthma, COVID-19, or healthy categories in real-time. This module uses parallel support-vector machines and a probabilistic voting mechanism to provide an interpretable alternative to deep networks.
3. We introduce a novel Generative Augmentation Module (GAM) comprising an ensemble of five distinct VAE architectures (including CVAE, VQ-VAE, and MLP-VAE variants) to synthesize realistic, disease-conditioned cough signals. To the best of our knowledge, this is the first application of such a multi-variant generative framework for respiratory acoustics. We specifically implement a cross-domain architecture where feature generation occurs in the time-frequency domain while final reconstruction is performed in the time-domain, ensuring that the synthetic samples are acoustically verifiable and clinically interpretable.
4. We demonstrate real-time, on-device operation on commodity Android hardware. All processing is performed locally, without transmitting audio or features to external servers, thereby preserving privacy and enabling deployment in connectivity-limited environments.

The remainder of this paper is organized as follows: [Section 2](#) details the framework architecture. [Section 3](#) describes the Cough Detection Module, [Section 4](#) presents the Disease Analysis Module, and [Section 5](#) introduces the VAE-based Generative Augmentation Module. [Section 6](#) discusses on-device deployment characteristics, [Section 7](#) discusses the limitations of this work, and [Section 8](#) provides conclusions and future directions.

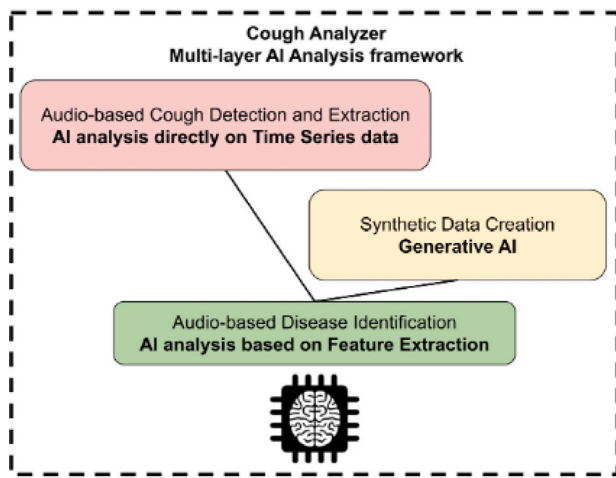
2. Framework overview

The proposed cough analysis framework ([Fig. 1](#)) is structured around three core modules that jointly encompass the offline training stage and the on-device inference stage. The first module, the **Cough Detection Module**, processes raw audio signals to isolate and segment complete cough episodes locally on the smartphone. The second module, the **Generative Augmentation Module**, employs several VAE variants to generate synthetic cough samples. It operates exclusively during training to enhance classifier robustness and is not deployed on the device. The third module, the **Disease Analysis Module**, evaluates features extracted from detected cough events to classify users into asthma, COVID-19, or healthy categories.

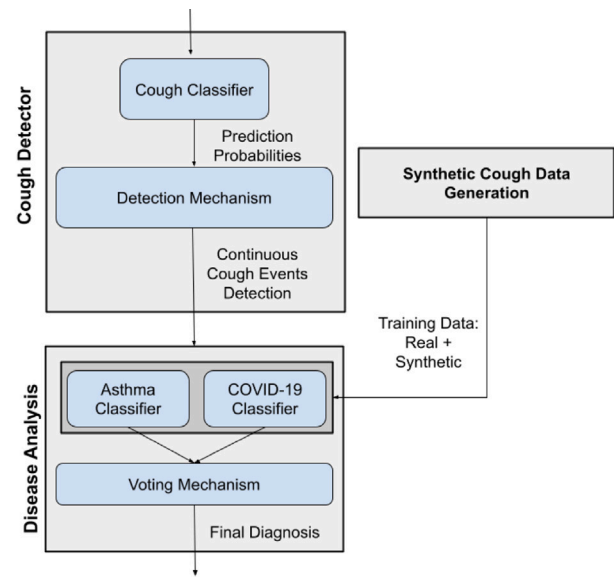
When integrated, the deployed modules (CDM and DAM) form a unified, smartphone-native AI framework that enables real-time and privacy-preserving passive respiratory monitoring using only the device’s built-in microphone. The offline generative module enhances model reliability during development.

3. Cough detection module

The *Cough Detection Module* (CDM) constitutes the first stage of the proposed cough analysis framework and is responsible for identifying cough sounds directly from continuous audio streams recorded by the smartphone. In contrast to approaches that detect only isolated cough bursts, our objective is to capture complete cough episodes. These



(a) High-level overview of the proposed cough analysis framework.



(b) Internal components overview within each module.

Fig. 1. Framework overview: (a) high-level representation of the proposed cough analysis framework, outlining its three core modules, and (b) internal components constituting each module.

episodes often consist of multiple temporally adjacent cough events and provide richer information for downstream disease analysis.

To achieve this, the CDM is organized into two main components. The first is a cough sound classification component, which implements an AI-based audio processing pipeline capable of discriminating between cough and non-cough acoustic segments. This pipeline encompasses feature extraction operations and a neural network classifier trained on short, overlapping audio frames. The second component is an event detection and episode extraction mechanism, which interprets the classifier's frame-level predictions over time and aggregates them into coherent cough events and multi-burst episodes.

In the following subsections, we detail (i) the dataset used to train the CDM, (ii) the architecture of the cough sound classifier, and (iii) the event detection and episode extraction mechanism. We then describe the evaluation protocol and report empirical performance results for both offline and on-device scenarios.

3.1. Related work

Early approaches to cough detection treated the task as a standard audio pattern-recognition problem based on hand-crafted features. Systems typically relied on Mel-frequency cepstral coefficients (MFCCs), short-time energy, Hu moments or non-negative matrix factorisation (NMF), combined with Gaussian mixture models (GMMs), k-nearest neighbours (KNN) or support-vector machines (SVMs). Reported performances included sensitivities of 80–89% and specificities above 90%, although positive predictive values were often limited and generalization across subjects remained a challenge [9]. These methods demonstrated feasibility; however, they were sensitive to background noise and recording variability.

Commercial ambulatory monitors such as the Leicester Cough Monitor (LCM), VitaloJAK and LEOSound established early standards for objective cough counting [10,11]. LCM uses MFCCs with a hidden Markov model backend, VitaloJAK employs free-field acoustic recordings, and LEOSound relies on wearable bioacoustic sensors [12]. While clinically adopted, these systems focus primarily on counting isolated coughs and provide limited temporal resolution for analysing complete cough episodes.

More recent work has examined segmentation algorithms aimed at reducing annotation requirements and improving boundary accuracy.

A boundary-regression approach based on a C-BiLSTM model demonstrated improved robustness to noise when estimating cough onsets and offsets [13]. Comparative analyses of amplitude- and hysteresis-based thresholding further showed that simple signal-level rules can offer stable segmentation for isolated events [14].

Deep learning now constitutes the predominant family of methods for cough detection. Convolutional recurrent neural networks trained on log-mel spectrograms have reported sensitivities and specificities exceeding 90% [15], while transformer-based models applied to spectro-temporal representations have achieved competitive performance under diverse acoustic conditions [16]. Machine-hearing representations combined with SVMs have also remained viable, obtaining an AUC of 0.91 in heterogeneous settings [17]. Although these methods yield strong detection accuracy, they generally require substantial computational resources or rely on large annotated corpora, which can limit their suitability for deployment on mobile devices.

3.2. Dataset

To construct a suitable training dataset for the cough sound classification component of the CDM, we combined subsets from three publicly available datasets. In total, 1448 audio recordings were collected from COUGHVID [18], COVID-19 Sounds [19], and Virufy [20]. These samples originated from manually verified, high-quality subsets of each dataset; however, they did not include time-aligned annotations indicating the precise locations of cough events within the recordings.

To extract cough-only segments, we employed a modified version of the segmentation algorithm proposed by Atmaja et al. [14]. The method uses a hysteresis-based energy thresholding mechanism to detect the onset and offset of candidate cough events. An illustration of the segmentation process is shown in Fig. 2.

Although effective in isolating cough-like acoustic patterns, the segmentation algorithm frequently misidentified terminal episode phases—often dominated by voiced or expiratory sounds—as separate cough events. As a result, some segments were not true coughs, despite exhibiting partial temporal or spectral similarity. To automatically refine these outputs, we introduced a semi-supervised K-Means clustering [21] procedure.

In this approach, an initial set of 50 high-confidence cough samples was used to seed the centroid representing the cough cluster, while

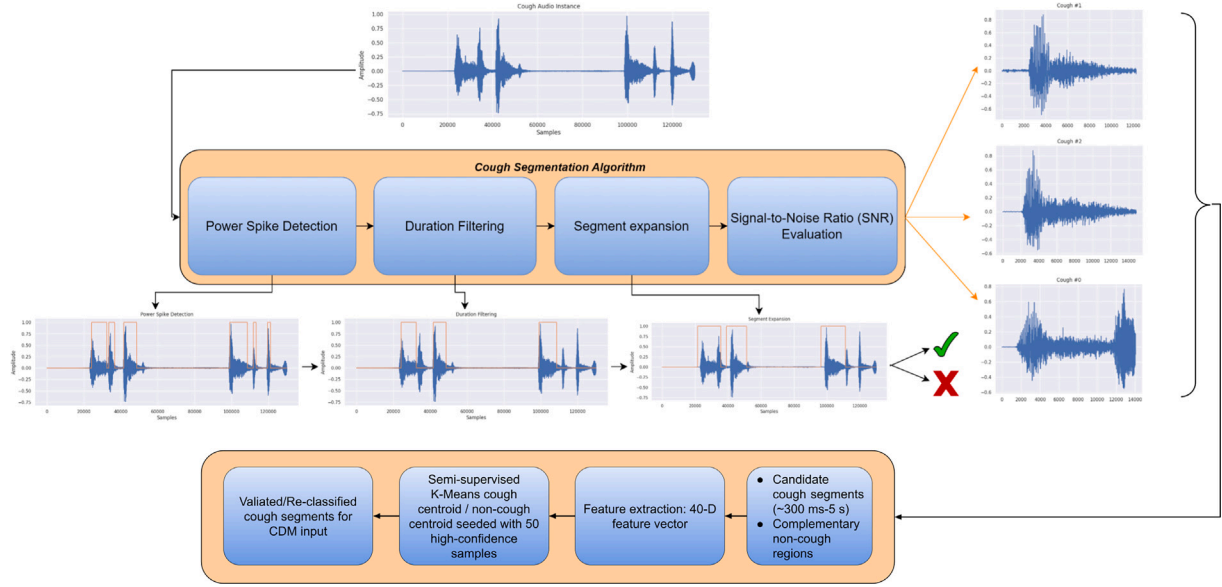


Fig. 2. Cough segmentation pipeline for extracting candidate events from continuous audio. Raw waveforms are resampled to 16 kHz mono; hysteresis thresholding on instantaneous power $x^2[n]$ uses an upper trigger $\tau_H = 2 \cdot \text{RMS}(x)$, a lower release $\tau_L = 0.1 \cdot \text{RMS}(x)$, a 10 ms release tolerance, a 200 ms minimum-duration filter, and 50 ms symmetric padding. Each variable-length candidate segment (~ 300 ms–5 s), together with the complementary non-cough portions of the waveform, is summarized into a 40-D feature vector (26-D log-mel energy, 4-D spectral-flux statistics, 4-D ZCR statistics, 6-D temporal-envelope descriptors) and refined by semi-supervised K-Means (Section 3.2), which reassigns mis-segmented voiced/expiratory spikes. Refined cough segments form the CDM cough class; non-cough samples are drawn from ESC-50. Validated segments are then windowed to 300 ms (4800 samples, 75% overlap) for the CDM 1D-CNN.

a second centroid was initialized using 50 high-confidence non-cough samples (e.g., clean speech or ambient sounds). Each segmented event was summarized into a 40-D feature vector spanning four descriptor families: 26-D log-mel energy (13 mel bands pooled as per-band mean and standard deviation), 4-D spectral-flux statistics (mean, standard deviation, maximum, slope), 4-D ZCR statistics (mean, standard deviation, rate-above-threshold, normalized zero-crossing count), and 6-D temporal-envelope descriptors (attack time, decay time, peak amplitude, RMS, duration, crest factor). The K-Means algorithm iteratively re-assigned samples to one of the two clusters. Segments assigned to the non-cough cluster were reclassified accordingly. After this refinement step, 1018 validated cough segments remained.

To construct the non-cough class, we selected a diverse subset of 599 recordings from the ESC-50 dataset [22]. From each recording, 2-second segments were extracted, covering a wide range of environmental and human-generated sounds, including speech, sneezing, clapping, bird-song, and other ambient noise. Combining these ESC-50 segments with the reclassified non-cough outputs resulted in a final set of 1242 non-cough segments, ranging mainly between 300 ms and 1.5 s, although several non-cough segments extended up to approximately 5 s.

Table 1 summarizes the composition of the CDM corpus, including the per-source breakdown of the 1448 raw audio recordings, the number of candidate cough events produced by hysteresis segmentation, the number of validated cough segments retained after semi-supervised K-Means refinement, the speaker-independent 70/20/10 split, and the DSP-augmentation yield on the cough training partition.

3.3. Cough classifier component

The cough classifier component is the core decision-making unit of the Cough Detection Module (CDM): it performs the cough-sound classification stage of the module (Fig. 3(a)), discriminating between cough and non-cough audio segments before any event-level processing takes place. The component comprises two parts that are applied in sequence: a data processing pipeline that transforms raw audio into standardized model inputs, and a lightweight one-dimensional convolutional neural network (1D-CNN) designed for efficient inference on edge devices. Each part is described in turn below.

3.3.1. Data processing

The data processing pipeline follows a sliding-window segmentation strategy, applied consistently during both training and deployment. Each audio recording is segmented into fixed-length windows of 300 ms (4800 samples at 16 kHz) with a 75% overlap (3600 samples). Shorter trailing segments that do not meet the window length requirement are discarded. This segmentation exposes short-term temporal structure and effectively increases the number of available training samples.

Following segmentation, the audio windows undergo filtering and audio normalization. Although we initially explored both high-pass and low-pass filtering, we found that high-pass filtering did not improve classification and was therefore omitted. Instead, a third-order low-pass Butterworth filter with a cut-off frequency of 5 kHz was applied to attenuate higher frequencies. Finally, each segment was amplitude-normalized to the $[-1, 1]$ range to ensure numerical stability during training.

3.3.2. Neural network architecture

A one-dimensional convolutional neural network (1D-CNN) was developed to balance representational capacity with the computational constraints of smartphone deployment (Fig. 4). The network receives inputs of shape $B \times 4800 \times 1$ corresponding to batches of raw waveform segments. The architecture begins with two consecutive one-dimensional convolutional layers with 64 filters and a kernel size of 3, each followed by ReLU activation. A one-dimensional max pooling layer with a stride of 3 reduces the temporal resolution while retaining salient features. This is followed by two additional one-dimensional convolutional layers with 128 filters to capture higher-level temporal patterns.

A one-dimensional global average pooling layer aggregates temporal features into a compact representation, followed by a spatial dropout layer (rate 0.5) used both for regularization during training and for Monte Carlo inference when uncertainty estimation is required. The classifier head consists of a flatten layer and a fully-connected layer with a softmax activation producing two output logits corresponding to the cough and non-cough classes.

Table 1

Composition of the Cough Detection Module (CDM) corpus. Top: per-source breakdown of the 1448 audio recordings used for cough extraction plus the 599 ESC-50 recordings used as non-cough material, with approximate aggregate duration, the number of candidate cough events produced by the hysteresis-based segmentation stage, and the number of validated cough segments retained after semi-supervised K-Means refinement; for ESC-50, the non-cough-segment count is reported after 2-second windowing. Middle: speaker-independent 70/20/10 train/validation/test partition of the 2260 validated segments. Bottom: cough-only DSP augmentation applied exclusively to the training split (additive white noise, $\sigma = 0.01$; pitch shift, ± 0.5 semitones) and the resulting class-balanced training totals.

Source	Recordings	Duration	Candidates (post-segm.)	Cough segments	Non-cough segments
COUGHVID [18]	700	4.5 h	1050	498	–
COVID-19 Sounds [19]	538	3.8 h	750	381	–
Virufy [20]	210	0.9 h	250	139	–
ESC-50 [22] (non-cough)	599	4.2 h	1242	–	1242
Total	2047	13.4 h	3292	1018	1242
Class	Train (70%)	Val (20%)	Test (10%)	Total	
Cough	713	204	101	1018	
Non-cough	869	249	124	1242	
Total	1582	453	225	2260	
DSP augmentation (cough-train only)			Samples added	Cough-train cumulative	
Real (no augmentation)			713	713	
+ Additive white noise ($\sigma = 0.01$)			78	791	
+ Pitch shift (± 0.5 semitones)			78	869	
Non-cough-train (reference)			–	869	

Note. All 5-fold cross-validation partitions were drawn exclusively from the 2260-segment corpus above; the 50 real-world pilot recordings used for event-level evaluation of the EDEM were collected independently of this corpus and never appeared in any training or validation fold.

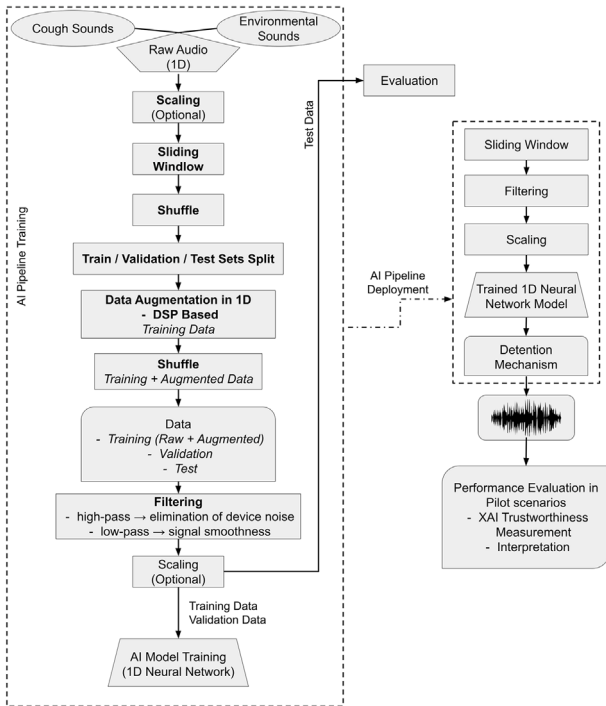


Fig. 3. Flowchart of the cough Detection Module, illustrating its two internal components: (a) cough sound classification and (b) event detection and extraction.

3.3.3. Training details

To mitigate class imbalance, augmentation was applied exclusively to cough-labeled training samples. Two digital signal processing (DSP) techniques were employed: (i) additive white noise with a magnitude factor of 0.01, and (ii) pitch shifting by a factor of 0.5. These augmentations were applied only to the training set to avoid distributional

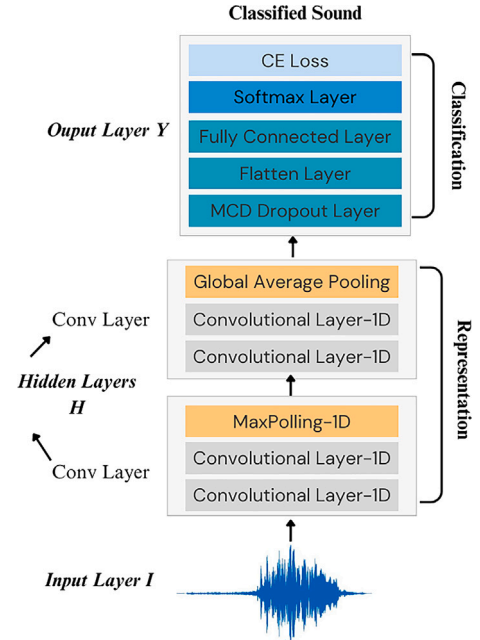


Fig. 4. Neural network architecture.

shift in validation and test data. In addition to augmentation, weighted random sampling was used during training to further balance class frequencies.

The model was optimized using the Adam optimizer with default momentum parameters ($\beta_1 = 0.9, \beta_2 = 0.999$). Training minimized the categorical cross-entropy loss. Early stopping was triggered based on the area under the precision-recall curve (PRC-AUC) on the validation set, which served as a more informative criterion than accuracy due to the remaining imbalance between classes. The best-performing model employed a learning rate of 0.005 with a Reduce-on-Plateau scheduler

that progressively decreased the learning rate down to 10^{-8} , a batch size of 256, and a maximum of 300 training epochs. Early stopping was applied when validation PRC-AUC failed to improve for 15 consecutive epochs.

3.4. Event detection and extraction mechanism

The *Event Detection and Extraction Mechanism (EDEM)* forms the second component of the Cough Detection Module and is responsible for transforming the classifier's frame-level probability outputs into coherent, continuous cough events, following the temporal event recognition and extraction approach introduced in Tzamalís et al.[23] and adapted here for acoustic cough signals. Its objective is to operate on audio streams longer than the classifier's window length and to reliably identify the temporal boundaries of cough episodes.

More specifically, as shown in Fig. 5, the EDEM receives the sequence of overlapping windows generated by the data processing pipeline and processes the corresponding classifier output probabilities through several stages. First, to improve robustness and quantify predictive uncertainty, we perform *Monte Carlo Dropout* inference with ten stochastic forward passes. Averaging these predictions produces a smooth probability matrix while retaining the epistemic uncertainty information.

Next, to obtain a continuous representation of class probabilities over time, each class-specific probability sequence is interpolated using cubic splines. The interpolated probability curves are then further smoothed with a low-pass filter to reduce local fluctuations and suppress noise.

Event extraction is subsequently performed through probability and duration thresholding. Probability values below a selected threshold (e.g., 50%) are suppressed, and only contiguous regions exceeding

a minimum duration (e.g., 200 ms) are retained. This step removes transient detections that do not correspond to physiologically plausible cough sounds, yielding well-defined cough episodes rather than isolated windows.

The final output of the EDEM consists of (i) the start and end timestamps of each detected event, and (ii) the corresponding cropped audio segments extracted directly from the input signal. These outputs provide structurally coherent cough episodes, enabling downstream disease analysis and on-device real-time monitoring. Fig. 6 visualizes the step-by-step execution process.

3.5. Experiments and evaluation

The evaluation of the Cough Detection Module was conducted in two stages. First, we assessed the cough classifier component at the window level. Second, we evaluated the Event Detection and Extraction Mechanism (EDEM) at the event level. To ensure a comprehensive and unbiased analysis, we employed a combination of Bayesian hyperparameter optimization, cross-validation, and pilot data testing.

3.5.1. Experimental setup

The dataset described in Section 3.2 was split into training, validation, and test sets in a 70/20/10% ratio. A speaker-independent partitioning strategy was used to prevent leakage of audio segments or subjects across splits. Hyperparameter optimization was performed using Bayesian optimization, where all data-processing and model-related parameters were jointly explored. For the cough classifier, the model selection criterion was the minimum validation loss. Each candidate model was evaluated under 5-fold cross-validation, and the mean and standard deviation across folds were recorded.

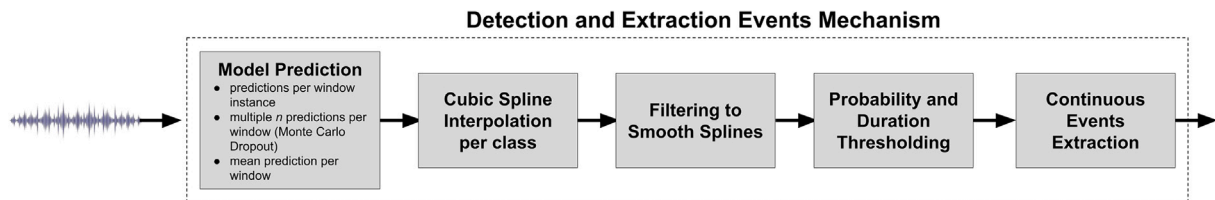


Fig. 5. Overview of the Event Detection and Extraction Mechanism (EDEM), which converts window-level probability estimates into continuous cough events.

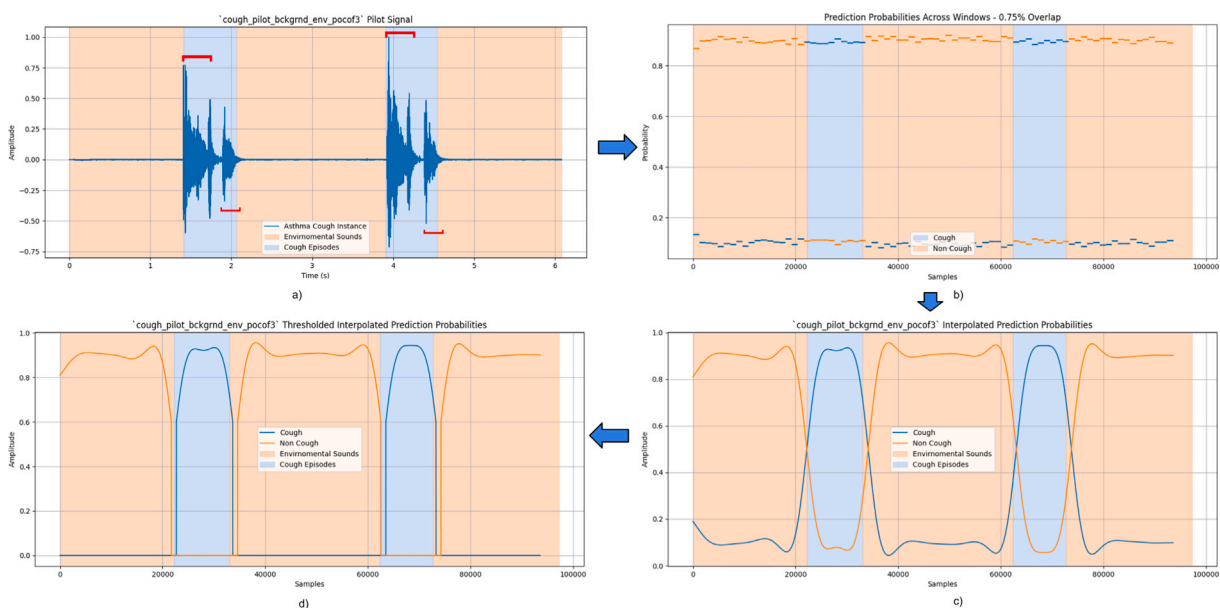


Fig. 6. Step-by-step visualization of the event detection and extraction mechanism (EDEM). Window-level classifier probabilities are interpolated, temporally smoothed, and thresholded to identify coherent cough events with well-defined temporal boundaries. The processing stages are shown sequentially across subfigures (a)–(d). Red horizontal brackets indicate individual cough events within a cough episode.

The hyperparameter search space included both signal-processing parameters (sampling rate, window size, overlap ratio, filter cut-off frequencies, filter orders, and normalization options) and neural network parameters (learning rate, type of dropout, and dropout rate). This joint exploration allowed the optimization procedure to consider interactions between processing choices and model architecture. This ensured that the resulting configuration was globally optimal for the end-to-end CDM pipeline.

Two main experiments were conducted: (i) a baseline model trained on the original dataset, and (ii) an augmented model where DSP-based augmentations were applied exclusively to cough-labeled training examples. Weighted random sampling was also used to mitigate class imbalance.

3.5.2. Classifier evaluation

The first evaluation layer focused on window-level cough vs. non-cough classification. Performance was quantified using loss, F1-score, and PRC-AUC on the held-out test set.

F1-score and PRC-AUC follow their standard definitions [24]. F1 is the harmonic mean of precision and recall, computed on window-level predictions with the cough class as the positive class; PRC-AUC summarizes precision–recall performance across the full decision-threshold range and is particularly informative under class imbalance because it is insensitive to the large number of true negatives contributed by the non-cough class.

Additional robustness analysis was conducted using *Monte Carlo (MC) Dropout* with $T = 10$ stochastic forward passes. For each input window $\mathbf{x}$, the predictive mean and variance are estimated as:

$$\hat{p}(\mathbf{x}) = \frac{1}{T} \sum_{i=1}^T f^{\tilde{\mathbf{w}}_i}(\mathbf{x}) \quad (1)$$

$$\widehat{\text{Var}}(\mathbf{x}) = \frac{1}{T} \sum_{i=1}^T \left[f^{\tilde{\mathbf{w}}_i}(\mathbf{x}) \right]^2 - \hat{p}(\mathbf{x})^2 \quad (2)$$

where $f^{\tilde{\mathbf{w}}_i}(\mathbf{x})$ is the model output under the i -th sampled dropout mask $\tilde{\mathbf{w}}_i$, and $\widehat{\text{Var}}(\mathbf{x})$ serves as a proxy for predictive uncertainty.

3.5.3. Event detection and extraction evaluation

The second evaluation layer examined the performance of the EDEM in identifying continuous cough events from long audio recordings. Event-level metrics were computed using manually annotated ground truths from a pilot dataset containing 50 real-world recordings of 20 s in length, with overlapping respiratory and environmental sounds.

Event detection performance was quantified using the Detection Ratio and the Recognition Error Rate. These metrics are derived from four event-matching outcomes based on the Intersection-over-Union (IoU) between detected segments and ground-truth cough boundaries.

Correct detections (C) correspond to events exceeding the IoU threshold. Deletions (D) arise when a ground-truth event is missed or has insufficient overlap. Insertions (I) indicate spurious detections with no IoU overlap. Substitutions (S) correspond to misclassified events failing to match the correct ground truth.

Formally, the Detection Ratio is defined as:

$$\text{Detection Ratio} = \frac{C}{D + C + S}, \quad (3)$$

and the Recognition Error Rate is:

$$\text{Recognition Error Rate} = \frac{D + I + S}{D + C + S} \quad (4)$$

During evaluation, all hyperparameters influencing event extraction—including probability threshold, minimum event duration, and low-pass filter order—were treated as tunable parameters. The optimal configuration used a probability threshold of 70%, a minimum duration of 300 ms, and a second-order smoothing filter.

3.5.4. Results and discussion

Table 2 summarizes the performance of the two main experimental conditions. DSP-based augmentation led to consistent improvements across both evaluation layers.

At the window level, the cough classifier achieved a mean F1-score of 92.0% (± 1.9) and a PRC-AUC of 0.94 (± 0.01) under five-fold cross-validation, compared to 89.0% (± 2.3) and 0.91 (± 0.02) for the baseline model, indicating that augmentation enhanced the model's ability to discriminate cough segments from non-cough acoustic events across all decision thresholds. Predictive uncertainty, estimated via Monte Carlo Dropout ($T = 10$ stochastic forward passes), also decreased from a mean variance of 0.014 (± 0.03) in the baseline to 0.009 (± 0.02) under DSP augmentation, suggesting that the augmented model produces more confident and consistent predictions. This gain propagated to the event level as well: the EDEM achieved a higher Detection Ratio (90.0% ± 3.4) and a lower Recognition Error Rate (21.2% ± 3.1) relative to the baseline configuration (84.0% ± 4.1 and 27.5% ± 3.8 , respectively), indicating the generalization and robustness capabilities of the AI pipeline. The relatively low standard deviations across folds and pilot recordings further demonstrate the stability of these improvements.

The absolute values of the classifier and EDEM metrics cannot be directly compared, as they correspond to fundamentally different tasks. The cough classifier operates on short, overlapping windows and performs a binary discrimination measured using standard metrics such as the F1-score and PRC-AUC. The PRC-AUC is particularly informative under class imbalance, as it evaluates precision–recall trade-offs across all thresholds rather than at a single operating point. The MC Dropout variance further complements these metrics by quantifying epistemic uncertainty, providing insight into model confidence beyond

Table 2

Performance of the Cough Detection Module across baseline and DSP-augmented experiments on the held-out test set. Window-level metrics (**F1-score**, **PRC-AUC**, and **MC Var.**) are computed on the cough/non-cough dataset using 5-fold cross-validation (mean $\pm$ standard deviation). **MC Var.** denotes the mean predictive variance estimated via Monte Carlo Dropout ($T = 10$ stochastic forward passes), serving as a proxy for model uncertainty. Event-level metrics (**Detection Ratio (DR)** and **Recognition Error Rate (RER)**) are computed on the manually annotated pilot dataset, averaged across recordings (mean $\pm$ standard deviation). Standard deviations in the event-level metrics reflect variability both in the EDEM and in the underlying cough classifier predictions. F1-score, DR, and RER are expressed as percentages; PRC-AUC is expressed in [0, 1]; MC Var. is a unitless uncertainty estimate.

Experiment	F1-score	PRC-AUC	MC Var.	DR	RER
Window-Level Classification on Cough/Non-cough Dataset, 5-fold CV (Cough Classifier Component)					
Baseline Cough vs. Non-cough	89.0% (± 2.3)	0.91 (± 0.02)	0.014 (± 0.03)	–	–
DSP Cough Augmentation	92.0% (± 1.9)	0.94 (± 0.01)	0.009 (± 0.02)	–	–
Event-Level Evaluation on Pilot Dataset (EDEM Component)					
Baseline Cough vs. Non-cough	–	–	–	84.0% (± 4.1)	27.5% (± 3.8)
DSP Cough Augmentation	–	–	–	90.0% (± 3.4)	21.2% (± 3.1)

point estimates. In contrast, the EDEM evaluates the module's ability to detect, aggregate, and segment cough episodes from continuous audio streams—a substantially more complex temporal task assessed using the Detection Ratio and Recognition Error Rate. Therefore, lower numerical values in EDEM metrics do not reflect inferior performance; rather, they capture temporal precision, boundary alignment quality, and resilience to real-world noise, all of which are inherently more challenging than frame-level classification.

Overall, these results demonstrate that the CDM provides accurate and confident frame-level predictions—as evidenced by the high F1-score, PRC-AUC, and reduced MC uncertainty—alongside reliable passive event-level detection, even under various acoustic conditions involving speech, environmental noise, and overlapping sounds.

4. Disease analysis module

The Disease Analysis Module (DAM) constitutes the second major stage of the proposed cough analysis framework and is responsible for identifying respiratory conditions based solely on cough events extracted by the Cough Detection Module. As illustrated in Fig. 1(b), the DAM operates directly on the segmented cough events produced by the CDM and transforms them into disease-specific predictions through a two-component architecture. The first component is a disease classifier comprising two Support Vector Machine (SVM) models trained independently for asthma and COVID-19. The second is a Voting Mechanism that integrates window-level predictions into a final episode-level outcome.

The choice of asthma and COVID-19 as target conditions reflects three considerations. First, both are covered by a single, publicly released smartphone-recorded dataset (Kuluozturk et al. [25]), which allows a consistent and reproducible experimental setup under uniform acquisition conditions. Second, they are among the most prevalent respiratory conditions worldwide and therefore represent high-impact targets for low-cost screening on commodity devices. Third, their cough acoustics differ substantially. Asthma is typically associated with wheezing and bronchoconstriction signatures, whereas COVID-19 coughs often exhibit dry, explosive patterns, providing a non-trivial yet tractable discrimination task [26].

The one-vs-rest SVM formulation of the disease classifier is itself easily extensible to additional disease categories: a new class is integrated by training one additional binary SVM on a class-specific dataset and registering its posterior in the Bayesian Posterior Fusion stage, with no retraining of the existing Asthma and COVID-19 classifiers, the gating constraint on taxonomy expansion is the availability of open, smartphone-recorded training data for such categories rather than classifier capacity (see the Future Work paragraph of Section 8). To prevent silent misclassification of out-of-taxonomy inputs (e.g., tuberculosis, bronchitis, chronic obstructive pulmonary disease), the Voting Mechanism (Section 4.4) operates over a four-label output space {Asthma, COVID-19, Healthy, *Uncertain*}: any input for which no disease-class posterior attains a calibrated probability threshold is emitted as *Uncertain* rather than coerced into one of the three trained categories; the decision rule is formalized in Section 4.4.1.

4.1. Related work

The classification of cough sounds into disease categories has been studied extensively, particularly during the COVID-19 pandemic. Early efforts relied on crowd-sourced datasets such as Coswara [27] and COUGHVID [18], together with conventional machine-learning models trained on handcrafted acoustic features. For example, Pal et al. [28] trained a shallow neural network to distinguish between COVID-19, asthma, bronchitis, and healthy coughs, reporting an accuracy of 96.83% on 328 recordings. Brown et al. [29] evaluated classical classifiers on cough and breath signals and obtained area-under-ROC

values slightly above 0.80. Bagad et al. [30] applied ResNet-18 to smartphone-recorded coughs from 3621 subjects, achieving an AUC of 0.72. Although these studies demonstrated the feasibility of acoustic disease screening, they were constrained by unbalanced datasets, heterogeneous recording conditions, and limited control over data quality.

Subsequent research has focused on deep learning and larger datasets. Celik et al. [31] proposed a hybrid InceptionFireNet–DeepConvNet model that achieved high accuracy on COUGHVID and Coswara datasets. Han et al. [32] introduced a VGGish-based system with separate detection and classification networks, reporting accuracies of 84.2–86.6% and F1 scores around 88% on expert-labeled smartphone coughs. These approaches illustrate that deep models can capture disease-specific patterns but typically require substantial computational resources and are not designed for on-device operation.

While our work focuses on cough-centric analysis, it aligns with a broader trend in biomedical signal processing toward lightweight, interpretable diagnostics. For instance, recent studies have explored other vocal biomarkers for COVID-19 using warped-frequency cepstral coefficients [33] and various machine-learning pipelines for speech-based respiratory screening [34]. Similar paradigms in feature engineering and metaheuristic optimization have also been successfully applied to neurological bio-signals, such as the use of bi-orthogonal wavelets [35] and advanced optimizers [36,37] for EEG-based screening. Although these works address different clinical taxonomies or signal modalities, they corroborate the feasibility of deploying resource-efficient, feature-engineered classifiers for automated medical screening.

Multi-class scenarios have also been explored. Kuluozturk et al. [25] developed the DKPNet41 architecture and demonstrated high performance on cough recordings from healthy subjects and patients with asthma, COVID-19 and heart failure. They reported an overall accuracy of 99.39%. This dataset is the one used in the present study for training and evaluating our Disease Analysis Module. While DKPNet41 shows strong results on the curated dataset, the authors do not report whether patient-level separation was enforced in their experimental splits. Nor did they examine the feasibility of deploying their model on mobile devices. Consequently, questions regarding generalization and real-world applicability remain open.

4.2. Dataset

For the disease classification task, we used the publicly available dataset introduced by Kuluozturk et al. [25]. This dataset contains smartphone-acquired cough recordings from individuals diagnosed with asthma, heart failure, COVID-19, or labeled as *Healthy* (i.e., without known respiratory or cardiac pathology). Participants were recruited from the Pulmonology and Cardiology Clinics of Firat University Hospital, Elazığ, Turkey.

The original recordings were stored in heterogeneous audio formats (MP3, OGG, MP4, M4A, MPEG) and sampled between 44.1 kHz and 48 kHz. As a first preprocessing step, all recordings were converted to mono-channel, 24-bit PCM WAV format and resampled to 22.5 kHz to ensure uniformity across the dataset. The curated dataset contains 787 asthma recordings, 696 normal recordings, 907 COVID-19 recordings, and 554 heart failure recordings, corresponding to 110 asthma subjects, 247 healthy subjects, and 241 COVID-19 subjects. Heart failure recordings were excluded from our study, as they do not fall within the scope of our respiratory-focused analysis.

Since the dataset does not include timestamps indicating where coughs occur within each recording, we applied the same cough segmentation procedure described in Section 3.2 to isolate individual cough episodes. This segmentation procedure increased the total number of training samples by approximately a factor of four, yielding a sufficiently large pool of cough events for downstream feature extraction and model training.

Table 3

Dataset composition of the disease analysis module (DAM) after cough segmentation. Heart-failure recordings from the Kuluozturk corpus [25] are excluded from this study. Duration reports aggregate audio length at the recording level (prior to segmentation). The windowing overlap differs across disease pipelines (asthma: 90%; COVID-19: 75%), as determined empirically during hyperparameter search; the healthy class reuses the windowing configuration of whichever SVM it is paired with under the one-vs.-rest training protocol. Train/test figures report the 80/20 speaker-independent split applied at the segment level prior to windowing.

Class	Recordings	Subjects	Duration	Cough segments	Train (80%)	Test (20%)	Window overlap
Asthma	787	110	0.9 h	3227	2582	645	90%
Healthy	696	247	0.6 h	2993	2394	599	90%/75% [†]
COVID-19	907	241	1.1 h	3809	3047	762	75%
Total	2390	598	2.6 h	10,029	8023	2006	–

[†] The healthy class is shared across both binary SVMs; its segments are re-windowed with the overlap of the paired disease pipeline during training.

Table 3 summarizes the DAM dataset composition after segmentation, along with the 80/20 train/test partition and the windowing overlap used by each disease pipeline.

4.3. Disease classifier component

In this section, we describe in detail the disease classifier component of the DAM, illustrated in Fig. 7. The component is composed of two parallel SVM-based classifiers—one for asthma identification and one for COVID-19 identification—each trained independently in a binary setting against the Healthy class. Although both classifiers operate on the same detected cough events, they employ distinct data processing pipelines and feature extraction procedures. These variations account for the differing acoustic signatures associated with each disease class.

4.3.1. Data processing

Once cough events are identified by the CDM, they are passed to dedicated data processing pipelines of each disease classifier. While

both pipelines share the same core structure, several hyperparameters differ due to the distinct acoustic signatures associated with asthma and COVID-19.

All segmented cough events are first resampled to 16 kHz to ensure consistency across recordings. Each event is then partitioned into 300 ms windows (4800 samples) using a sliding window approach. The asthma pipeline achieved optimal performance using a 90% overlap (4320 samples), whereas the COVID-19 pipeline performed best with a 75% overlap (3600 samples). Following segmentation, windowed samples are shuffled and split into training and test sets with an 80/20 ratio.

Both pipelines incorporate filtering to reduce noise and highlight medically relevant spectral content. The Asthma pipeline applies a third-order high-pass filter at 50 Hz followed by a third-order low-pass filter at 4 kHz. The COVID pipeline applies the same filter configuration but includes an additional audio-normalization step that rescales raw input waveforms to the $[-1, 1]$ range before segmentation. These variations were determined empirically through extensive grid-based exploration.

The feature extraction step also differs between the two disease pipelines. For asthma classification, the most informative representation consisted of the first 20 mel-frequency cepstral coefficients (MFCCs), aggregated by computing the mean across each time frame. All cepstral features were derived from a short-time Fourier transform (STFT) computed using a 512-point FFT and a hop length of 128 samples (8 ms at 16 kHz), with a Hann window.

For COVID-19 identification, combining cepstral and spectral power features provided superior discriminative performance. The COVID pipeline therefore extracts the first 13 MFCCs (mean aggregated) together with power spectral density (PSD) values computed from the same STFT configuration over six absolute frequency bands: 0–200 Hz, 500–650 Hz, 950–1150 Hz, 1400–1800 Hz, 2300–2400 Hz, and 2850–2950 Hz. These bands were identified during exploratory analysis as regions where COVID-positive coughs exhibit distinct spectral signatures.

Feature scaling also differs across pipelines. The Asthma pipeline applies a robust scaler with a quantile range of 0.3–0.7 to mitigate the influence of outliers and heavy-tailed distributions. In contrast, the COVID pipeline uses standardization (z-scoring), which stabilizes the classifier's decision boundaries given the higher variability of the combined MFCC + PSD feature space.

4.3.2. Disease classifiers

The disease classifier component consists of two parallel binary models: an *Asthma vs. Healthy* classifier (hereafter the *Asthma classifier*) and a *COVID-19 vs. Healthy* classifier (hereafter the *COVID classifier*). Although both models operate on the same windowed cough events at inference time, each is trained independently using its own disease-specific data processing pipeline and corresponding feature extraction method.

Both classifiers are implemented as SVMs with radial basis function (RBF) kernels, selected for their suitability in small-to-medium-sized

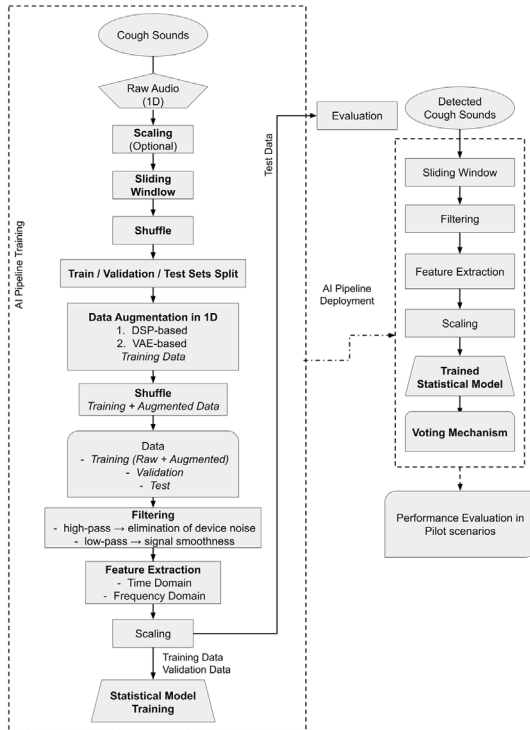


Fig. 7. Detailed internal structure of the disease analysis module, including data processing, feature extraction, parallel disease classifiers, and the voting mechanism.

biomedical audio datasets, their interpretability, and their low computational cost during on-device inference. The Asthma classifier uses an RBF kernel with $C = 100$ and $\gamma = \text{scale}$, while the COVID classifier employs an RBF kernel with $C = 100$ and $\gamma = 0.1$. Probability estimates are enabled for both models, and the one-vs.-rest (OVR) decision function is used to compute class-specific posterior probabilities. These probabilistic outputs are subsequently fused by the *Voting Mechanism* to obtain the final episode-level prediction, as described next.

4.4. Voting mechanism

The Voting Mechanism is the DAM's final decision stage: it consumes the per-window posterior probabilities emitted by the Asthma and COVID-19 SVMs and returns a single episode-level label drawn from the four-element output space {Asthma, COVID-19, Healthy, *Uncertain*}. The first three labels are determined by the Bayesian Posterior Fusion rule described below; the *Uncertain* label is emitted by the abstain rule of Section 4.4.1 whenever no class attains a calibrated posterior threshold.

A naïve approach to combining classifier outputs would assign equal weight to both SVMs. However, this ignores a critical observation: classifier reliability varies across cough events. For a given cough, one classifier may produce highly consistent probability estimates across its temporal windows, while the other may exhibit substantial fluctuation. Intuitively, consistent predictions indicate confidence, suggesting that the classifier has captured a genuine discriminative pattern instead of responding to random noise.

Bayesian model averaging formalizes this intuition by treating each classifier as a probabilistic model and weighting its contribution according to how well its outputs conform to an expected distribution. Classifiers producing tightly clustered, consistent outputs receive higher weight, while those with scattered, inconsistent outputs are down-weighted. Crucially, this weighting is computed *per-sample*, allowing the fusion mechanism to adapt to the specific characteristics of each cough event.

Formally, when a cough event is segmented into W windows, each window w yields probability estimates $p_{c,w}^{(d)}$ from classifier $d \in \{\text{Asthma}, \text{COVID}\}$ for class $c \in \{\text{Disease}, \text{Healthy}\}$. Let $\mathcal{M}_d$ denote the probabilistic model corresponding to classifier d , and let $\mathcal{D} = \{p_{\text{Disease},w}^{(d)}\}_{w=1}^W$ denote the observed window outputs.

The posterior model weight is computed via Bayes' theorem:

$$\pi_d \triangleq P(\mathcal{M}_d | \mathcal{D}) = \frac{P(\mathcal{D} | \mathcal{M}_d) P(\mathcal{M}_d)}{\sum_{d'} P(\mathcal{D} | \mathcal{M}_{d'}) P(\mathcal{M}_{d'})}, \quad (5)$$

where $P(\mathcal{M}_d)$ is a prior over models, set to uniform in the absence of domain knowledge.

The marginal likelihood $P(\mathcal{D} | \mathcal{M}_d)$ quantifies how well the observed outputs fit classifier d 's expected behavior. Assuming window outputs are approximately Gaussian-distributed around a classifier-specific mean:

$$P(\mathcal{D} | \mathcal{M}_d) = \prod_{w=1}^W \mathcal{N}(p_{\text{Disease},w}^{(d)} | \mu^{(d)}, (\sigma^{(d)})^2). \quad (6)$$

The fusion mechanism requires no additional training beyond the two SVM classifiers. After the classifiers are trained independently, the hyperparameters $\mu^{(d)}$ and $\sigma^{(d)}$ are estimated as the sample mean and standard deviation of each classifier's disease-class probability outputs computed on the training set:

$$\mu^{(d)} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} p_{\text{Disease},t}^{(d)}, \quad (7)$$

$$\sigma^{(d)} = \sqrt{\frac{1}{|\mathcal{T}| - 1} \sum_{t \in \mathcal{T}} (p_{\text{Disease},t}^{(d)} - \mu^{(d)})^2}, \quad (8)$$

where $\mathcal{T}$ denotes the set of training windows. At inference time, these statistics serve as the reference distribution against which cross-window

consistency is evaluated. When window outputs cluster tightly (low empirical variance relative to $\sigma^{(d)}$), the marginal likelihood is large; when they scatter widely, it diminishes. The model weights π_d thus serve as adaptive, sample-specific confidence measures.

The window-averaged disease probability for each classifier is:

$$\bar{p}_{\text{Disease}}^{(d)} = \frac{1}{W} \sum_{w=1}^W p_{\text{Disease},w}^{(d)}. \quad (9)$$

The three-class posterior probabilities are then constructed by weighting each classifier's contribution according to its inferred reliability:

$$P_{\text{Asthma}} = \pi_{\text{Asthma}} \cdot \bar{p}_{\text{Disease}}^{(\text{Asthma})}, \quad (10)$$

$$P_{\text{COVID}} = \pi_{\text{COVID}} \cdot \bar{p}_{\text{Disease}}^{(\text{COVID})}, \quad (11)$$

$$P_{\text{Healthy}} = 1 - P_{\text{Asthma}} - P_{\text{COVID}}. \quad (12)$$

The final prediction is obtained via the maximum a posteriori (MAP) decision:

$$\hat{y} = \arg \max_{c \in \{\text{Asthma}, \text{COVID}, \text{Healthy}\}} P_c. \quad (13)$$

4.4.1. Abstain option for out-of-taxonomy inputs

The three-class MAP decision in Eq. (13) implicitly assumes that every input cough belongs to one of the three trained categories. In an open-world deployment this assumption is violated whenever a user presents with a respiratory condition outside the trained taxonomy (e.g., tuberculosis, pneumonia, chronic obstructive pulmonary disease). To prevent such inputs from being silently coerced into one of the three labels, we augment the decision rule with an explicit abstain option:

$$\hat{y} = \begin{cases} \arg \max_{c \in \{\text{Asthma}, \text{COVID}, \text{Healthy}\}} P_c, & \text{if } \max_c P_c \geq \tau, \\ \text{Uncertain}, & \text{otherwise,} \end{cases} \quad (14)$$

where $\tau \in [0, 1]$ is a posterior-probability threshold. Inputs for which no class attains a probability of at least τ are not assigned a label; instead they are surfaced to the user as *Uncertain*. The threshold τ trades *coverage* (the fraction of inputs that receive a confident label) against *precision on covered inputs*, and is selected on the held-out validation folds to maximize covered-set F1 subject to a coverage floor, over the three-way decision surface.

Intuitively, lower-bounding the maximum class posterior by τ is a simple yet effective way to filter out out-of-taxonomy inputs, provided the two binary SVMs have been trained on representative samples of their respective disease-signature acoustics: a cough whose spectral and cepstral features lie outside both training distributions—e.g., tuberculosis, pneumonia, or COPD—typically drives both the Asthma and COVID-19 per-class posteriors to diffuse, small values, so no class clears τ and the Voting Mechanism returns *Uncertain* rather than silently coercing the input onto one of the three trained labels. The rule therefore leverages, rather than bypasses, the calibration of the underlying classifiers.

4.5. Experiments and evaluation

We evaluated the Disease Analysis Module by assessing the performance of its two SVM-based disease classifiers and the final Voting Mechanism. Experiments were conducted under two configurations. The baseline setup used only the original segmented cough events. The DSP-augmented setup expanded the training set via white-noise injection (factor 0.01) and pitch shifting (0.5 semitone). These augmentations were selected to increase acoustic variability while preserving disease-related acoustic properties.

All experiments were performed using stratified 5-fold cross-validation with strict speaker independence to avoid subject leakage. For each fold, two new SVM models (Asthma vs. Healthy and COVID-19 vs. Healthy) were trained on four folds and evaluated on the held-out fold. All reported metrics represent the mean and standard deviation across the five folds.

4.5.1. Classifier evaluation

Each disease classifier was evaluated independently in its binary configuration. Window-level predictions from the Asthma and COVID-19 classifiers were assessed using specificity, recall (sensitivity), and F1-score, all computed under their standard definitions [24], with the disease class as the positive class. These metrics quantify how well each classifier distinguishes disease-related cough windows from Healthy cough windows using the data processing pipelines described in Section 4.3.1. Because the two classifiers are trained independently, each produces its own set of classification statistics.

4.5.2. Voting mechanism evaluation

The Voting Mechanism is evaluated under both decision rules introduced in Section 4.4: the unfiltered three-class MAP rule of Eq. (13) and the abstain-augmented rule of Eq. (14). Under the unfiltered rule the DAM behaves as a standard three-class classifier over {Asthma, COVID-19, Healthy}, and we report per-class precision, recall, and F1-score for each of the three fusion strategies described in Section 4.4 (Uniform Averaging, Maximum Confidence, and Bayesian Posterior Fusion). These metrics are reported in Table 4 and characterize the intrinsic discriminative power of the DAM independently of the abstain mechanism.

Under the abstain-augmented rule the Voting Mechanism can return *Uncertain* whenever no class attains the posterior threshold τ . Two operating-point metrics that only make sense in the four-label regime are therefore introduced: *coverage*, the fraction of inputs on which the system commits to one of the three disease labels (i.e., does not abstain),

Table 4

Evaluation metrics (mean $\pm$ std) for the disease analysis module. Top: binary classifier performance. Bottom: three-class voting mechanism performance comparing fusion strategies. Best values per metric are highlighted in bold. Baseline trains on the real 80% partitions of Table 3 (Asthma SVM: 2582 Asthma vs. 2394 Healthy training segments; COVID SVM: 3047 COVID-19 vs. 2394 Healthy); DSP-Augmented applies one perturbation per real segment—additive white noise ($\sigma = 0.01$) or pitch shifting (± 0.5 semitone), drawn uniformly at random—yielding training compositions of 5164/4788 (Asthma SVM) and 6094/4788 (COVID SVM). No non-cough material is included in either SVM's training partition; the DAM operates exclusively on cough events already gated by the CDM's event detection and extraction mechanism (Section 3).

Model	Class	Specificity (%)	Recall (%)	F1-score (%)
Asthma Classifier (Binary: Asthma vs Healthy)				
Baseline	Asthma	74.8 $\pm$ 2.4	77.1 $\pm$ 2.2	75.9 $\pm$ 2.0
	Healthy	77.6 $\pm$ 2.1	76.4 $\pm$ 2.0	77.0 $\pm$ 1.9
DSP-Augmented	Asthma	76.3 $\pm$ 2.2	78.3 $\pm$ 2.0	77.0 $\pm$ 1.7
	Healthy	78.7 $\pm$ 2.0	77.8 $\pm$ 1.9	78.2 $\pm$ 1.8
COVID Classifier (Binary: COVID-19 vs Healthy)				
Baseline	COVID-19	73.9 $\pm$ 2.6	77.1 $\pm$ 2.4	75.5 $\pm$ 2.1
	Healthy	77.5 $\pm$ 2.3	76.0 $\pm$ 2.1	76.7 $\pm$ 2.0
DSP-Augmented	COVID-19	75.1 $\pm$ 2.4	78.2 $\pm$ 2.3	76.4 $\pm$ 1.9
	Healthy	78.6 $\pm$ 2.1	77.3 $\pm$ 1.9	77.9 $\pm$ 1.8
Voting Mechanism (3-Class: Asthma, COVID-19, Healthy) — DSP-Augmented				
Uniform Averaging	Asthma	70.1 $\pm$ 2.9	68.2 $\pm$ 2.7	69.1 $\pm$ 2.6
	COVID-19	68.8 $\pm$ 2.8	66.9 $\pm$ 2.7	67.8 $\pm$ 2.6
Maximum Confidence	Healthy	78.9 $\pm$ 2.0	81.7 $\pm$ 1.9	80.3 $\pm$ 1.9
	Asthma	71.4 $\pm$ 2.7	69.5 $\pm$ 2.5	70.4 $\pm$ 2.5
Bayesian Posterior Fusion	COVID-19	70.2 $\pm$ 2.6	68.1 $\pm$ 2.5	69.1 $\pm$ 2.4
	Healthy	79.8 $\pm$ 1.9	82.6 $\pm$ 1.8	81.2 $\pm$ 1.8
	Asthma	73.2 $\pm$ 2.6	71.0 $\pm$ 2.4	72.1 $\pm$ 2.3
	COVID-19	71.9 $\pm$ 2.5	69.8 $\pm$ 2.4	70.8 $\pm$ 2.3
	Healthy	81.4 $\pm$ 1.8	84.2 $\pm$ 1.6	82.7 $\pm$ 1.7

and *covered-set F1*, the macro-averaged F1 over {Asthma, COVID-19, Healthy} computed on covered inputs only. Table 5 reports the full coverage–F1 trade-off as a function of τ under the best fusion strategy identified in Table 4, from which the deployed operating point τ^* is selected.

The abstain mechanism re-uses the existing Bayesian Posterior Fusion outputs, adds no training overhead, and remains fully compatible with on-device inference. We emphasize that it is a lightweight screening safeguard rather than a substitute for a dedicated open-set recognizer; rigorous open-set recognition—which would require labeled out-of-distribution cough data recorded on commodity-smartphones—is discussed as future work in Section 8.

4.5.3. Results and discussion

Table 4 reports DAM performance under the unfiltered MAP decision of Eq. (13) in two complementary blocks. The top block evaluates each SVM classifier in its binary configuration: both the Asthma-vs-Healthy and COVID-19-vs-Healthy classifiers achieve stable behavior across cross-validation folds, with F1-scores in the mid-70% range for both the disease and Healthy classes, and DSP-based augmentation yielding approximately a +1.1% absolute F1 improvement for each disease class.

The bottom block evaluates the three-class Voting Mechanism under each of the three fusion strategies introduced in Section 4.4, all using the unfiltered decision rule of Eq. (13). Bayesian Posterior Fusion outperforms both Uniform Averaging and Maximum Confidence across all three classes, with the largest gains on the two disease classes (approximately 2–3% absolute improvement in F1-score over Uniform Averaging), confirming that adaptive weighting based on per-window classifier consistency is preferable to naïve fusion. As expected, the three-class F1-scores of the disease classes are slightly lower than those of the corresponding binary classifiers, because the final decision now coordinates two independently trained models; the Bayesian formulation mitigates this effect by up-weighting the more confident classifier, and the Healthy class benefits from the conservative nature of the fusion mechanism, attaining the highest specificity and recall among the three classes.

Having identified DSP-augmented Bayesian Posterior Fusion as the best fusion strategy under the unfiltered rule, we next select the deployed operating point τ^* of the abstain-augmented rule of Eq. (14) by sweeping $\tau \in [0.50, 0.90]$ on this configuration and measuring, at each operating point, the coverage and macro-averaged covered-set F1 on the held-out validation folds. Table 5 summarizes the sweep: coverage decreases monotonically with τ , while covered-set F1 increases from 75.3% at $\tau = 0.50$ (coverage 99.4%; effectively equivalent to the unfiltered three-class MAP decision) to 93.4% at $\tau = 0.90$ (coverage 43.2%). We adopt $\tau^* = 0.80$ as the default operating point, selected on the criterion

Table 5

Coverage–F1 trade-off of the abstain rule (Eq. (14)) across posterior-probability thresholds τ , measured on the held-out validation folds of the three-class Voting Mechanism under the DSP-augmented Bayesian Posterior Fusion configuration. Coverage is the fraction of inputs that receive a confident label; covered F1 is the macro-averaged F1 over the three classes computed on covered inputs only. The selected default operating point $\tau^* = 0.80$ is highlighted in bold.

τ	Coverage (%)	Covered F1 (%)
0.50	99.4	75.3
0.60	93.8	79.1
0.70	84.6	83.2
0.75	77.1	85.8
0.80*	68.5	88.4
0.85	57.3	90.9
0.90	43.2	93.4

of maximal covered-set F1 subject to coverage $\geq 65\%$. At τ^* the system attains 88.4% covered-set F1 with 68.5% coverage: a +13.1% absolute F1 gain on covered inputs over the unfiltered $\tau = 0$ baseline, achieved by abstaining on approximately 31.5% of inputs and recommending clinical evaluation in their place. τ^* is exposed as a configuration parameter in the deployed application so that sites with different precision–coverage preferences can retune the operating point without model retraining.

Overall, the DAM demonstrates stable behavior across folds and modest improvements over traditional DSP-based augmentation. Nevertheless, the observed performance levels also reflect a fundamental limitation of cough-based disease classification: the scarcity and heterogeneity of labeled cough recordings for specific respiratory conditions. In particular, the relatively limited number of disease-positive cough samples constrains the ability of discriminative models to fully capture the variability of pathological cough patterns. To address this limitation, we introduce a generative data augmentation strategy designed to enrich the training distribution with realistic synthetic cough representations.

5. VAE augmentation module

Deep generative modeling has recently gained substantial attention as an effective strategy for mitigating data scarcity, improving class balance, and enhancing robustness in medical machine learning systems. These challenges are particularly pronounced in respiratory acoustics, where disease-specific cough recordings are limited, heterogeneous, and often collected under uncontrolled conditions. In our dataset (Section 4.2), this scarcity is most evident for asthma- and COVID-19-related cough sounds, creating a strong motivation for principled generative augmentation.

To address this limitation, we develop and evaluate a comprehensive VAE-based augmentation module capable of synthesizing realistic cough audio associated with three target classes: asthma, COVID-19, and healthy controls. The following subsections first outline related work, then describe our methodological framework, and finally present an extensive evaluation of generative fidelity and downstream utility.

5.1. Related work

The scarcity of annotated samples, combined with class imbalance is a common problem in medical datasets and often leads to poor generalization and reduced model reliability. To address these issues, audio data augmentation is commonly employed to artificially increase dataset size and variability [38–40]. Conventional augmentation methods rely on digital signal processing (DSP) techniques such as pitch shifting, noise injection, time warping, and spectrogram masking [40]. While these methods are simple to implement, they introduce only limited variability and may distort the semantic properties of cough sounds. In some cases, excessive augmentation can lead to distributional mismatch or unintended artifacts, degrading model performance [41].

Despite the growing interest in respiratory acoustics, research on generating realistic synthetic cough sounds remains limited. Existing attempts primarily rely on waveform-based generative adversarial networks. For example, CoughGAN [42] and Yella et al. [43] employ WaveGAN [44] to synthesize cough audio for binary COVID-19 classification. However, these approaches evaluate synthetic data only through downstream classifier performance and do not assess perceptual quality or distributional similarity to real cough signals. Furthermore, their focus on a single waveform generator highlights a broader challenge: modeling raw audio in the 1D time domain is inherently complex due to its high dimensionality and fine temporal granularity. This complexity often leads to instability during training and limited interpretability [45].

5.2. Methodology

This section presents the proposed VAE-based generative module for synthesizing class-conditioned cough sounds. We investigate five variational architectures that represent the major families of VAE models

used in audio generation: a fully connected MLP-VAE, a Convolutional VAE (CVAE), a Conditional CVAE (CCVAE) [46], a β -VAE [47] emphasizing disentangled representations, and the Vector Quantized VAE (VQ-VAE) [48]. Each model is trained on real cough audio and evaluated under multiple protocols to quantify both generative fidelity and practical utility for downstream disease classification.

We adopt a spectrogram-based generation approach, in which Mel-spectrograms serve as the target representation for learning the latent distribution of cough acoustics. This choice enables stable training, leverages convolutional inductive biases, and preserves the time-frequency structure characteristic of pathological cough patterns. After generation, spectrograms are converted back to the time domain to allow objective similarity assessments and integration into the Disease Analysis Module.

Generative performance is evaluated using three widely accepted audio similarity metrics: Fréchet Audio Distance (FAD) [49], Cross-Correlation (CC), and Mel Cepstral Distortion (MCD). Additionally, we evaluate the clinical relevance of synthetic samples through two classification-oriented protocols, Train-on-Real-Test-on-Synthetic (TRTS) and Train-on-Synthetic-Test-on-Real (TSTR), as well as hybrid training setups where generated samples are used to balance minority disease classes. The overall VAE augmentation framework is illustrated in Fig. 8.

5.2.1. Audio processing pipeline

The audio processing pipeline is responsible for preparing the cough recordings for training the generative models. To standardize input lengths, each audio recording is segmented using a sliding window of 600 ms, with a 50% overlap. Segments shorter than this duration are zero-padded symmetrically to preserve the central region of the cough event and maintain temporal consistency. The dataset is then split into training and testing subsets using an 80%–20% ratio. Only the training set is used for feature extraction and generative model training, while the test set is reserved for evaluating the quality of generated samples and the performance of classification models.

Each training audio segment is converted into a time-frequency representation using 128 Mel filter banks, a frame size of 512 samples, and a hop length of 128 samples. The resulting Mel-spectrograms have dimensions of 128×102 (frequency $\times$ time) and are transformed from the power scale to the decibel (dB) domain. During this transformation, all spectrograms are normalized relative to their maximum power and clipped at a fixed lower bound, resulting in a consistent amplitude range of -80 dB to 0 dB across all samples.

To prepare the data for input to the VAE architectures, a Min–Max normalization is applied to map the spectrogram values from $[-80, 0]$ dB to the range $[0, 1]$. This normalized representation is used both during training and when decoding generated spectrograms for inversion back to the time domain. Additionally, for compatibility with convolutional layers, the time axis of each spectrogram is zero-padded to a fixed size of 128, yielding a final input shape of $N \times 128 \times 128$, where N denotes the number of training samples.

Table 6 reports the number of real cough samples fed into each class-conditional VAE after 600 ms windowing (50% overlap, symmetric zero-padding for shorter segments), together with the size of the per-class synthetic pool generated by each VAE architecture and the subset retained for acoustic-quality evaluation (Section 5.3). The pool is sized to support the principal $3.5\times$ per-class augmentation ratio adopted in Table 8 as well as the full magnitude-sensitivity sweep up to $6.0\times$ reported in Table 11, with 5-fold disjoint draws across folds. All samples, both real and synthetic, are 600 ms long by construction, so the aggregate duration is reported in a dedicated column alongside the sample counts.

5.2.2. Variational autoencoders

Variational Autoencoders are generative models composed of an encoder and a decoder trained jointly by maximising the Evidence Lower

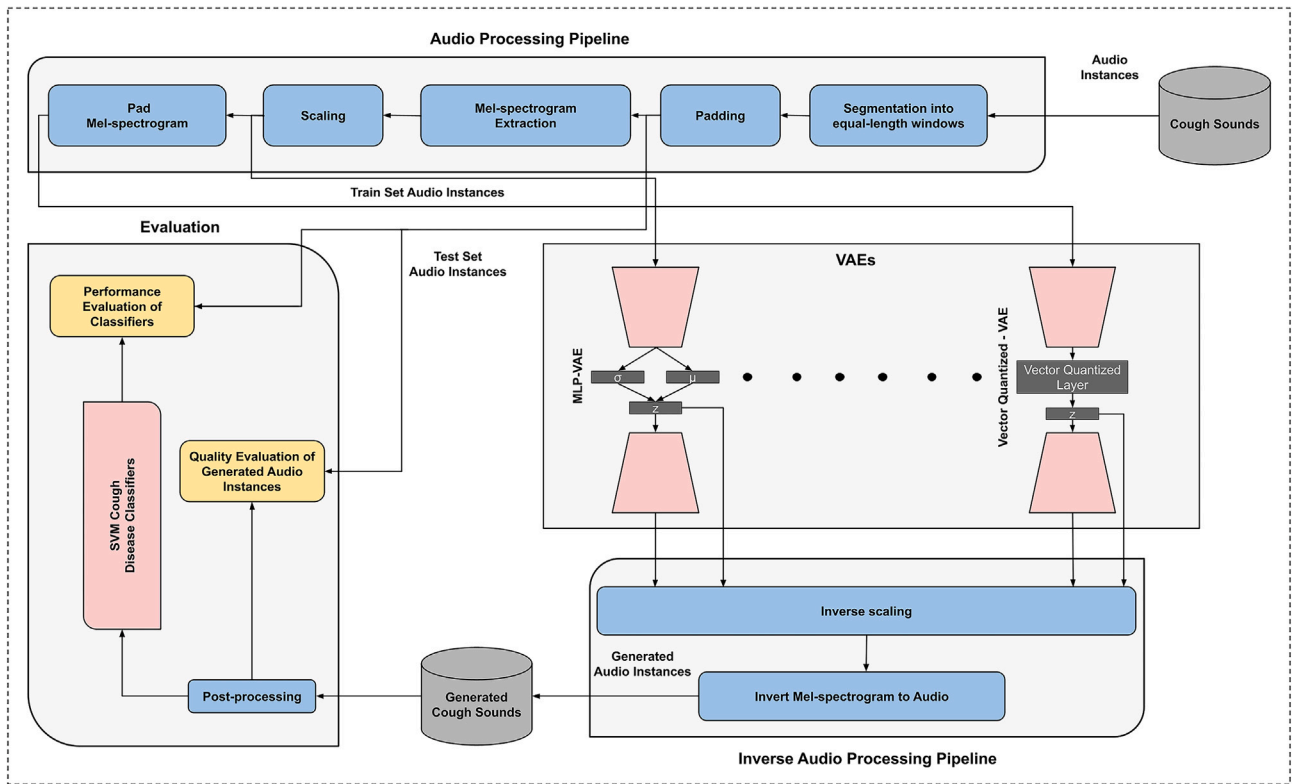


Fig. 8. Frequency domain cough sound generation VAE module.

Table 6

VAE training composition per class. Real samples are drawn from the 80% training split of the DAM corpus (Table 3) after 600 ms windowing with 50% overlap; each sample is a fixed-length 600 ms window. Each VAE architecture (MLP-VAE, CVAE, CCVAE, β -VAE, VQ-VAE) is trained independently per class and generates a per-class synthetic pool of 16,000 samples, sized to support both the principal 3.5 $\times$ augmentation ratio (5-fold draws of 6455/7620/5985 for Asthma/COVID-19/Healthy) and the full sensitivity sweep up to 6.0 $\times$ reported in Table 11; 200 per class are randomly drawn for acoustic-quality evaluation (FAD/CC/MCD/TRTS/TSTR). The duration column aggregates the total audio time across the real training pool and the generated synthetic pool per VAE architecture.

Class	Real samples	Real duration	Synthetic per VAE	Synthetic duration	Acoustic eval.
Asthma	2582	25.8 min	16,000	160.0 min	200
COVID-19	3047	30.5 min	16,000	160.0 min	200
Healthy	2394	23.9 min	16,000	160.0 min	200
Total	8023	80.2 min	48,000	480.0 min	600

Bound (ELBO) on the marginal log-likelihood of the observed data; we refer the reader to the foundational work of Kingma and Welling [50] for the formal definition of the ELBO, the Gaussian-posterior parameterisation of the approximate posterior $q_{\phi}(z | x)$, and the reparameterisation trick used to back-propagate through stochastic samples. In this study, we evaluate five VAE variants to explore their ability to generate realistic respiratory audio: the Multilayer Perceptron VAE (MLP-VAE), the Convolutional VAE (CVAE) [50], the Conditional CVAE (CCVAE) [46], the β -VAE [47], and the Vector Quantized VAE (VQ-VAE) [48]. The following subsections describe the specific encoder/decoder architectures, latent-space dimensions, and training hyperparameters we selected for each variant; the underlying generative formulations are standard and the reader is referred to the cited works for their derivations.

Multilayer perceptron VAE (MLP-VAE). The MLP-VAE [50] employs multilayer perceptrons for both the encoder and decoder. The input Mel-spectrogram of shape 128×128 is flattened into a 1D vector before being passed to the encoder. The encoder consists of four fully connected

(dense) layers with 256, 128, 64, and 32 units, respectively. These layers are followed by two additional dense layers that output the mean μ and log variance $\log \sigma^2$, with a latent space dimension of 128. The use of $\log \sigma^2$ instead of σ improves numerical stability during training.

The decoder mirrors the encoder's structure, starting with a dense layer that receives the sampled latent vector. A reshape layer transforms the latent vector into a suitable input shape for the subsequent upsampling layers. Four additional dense layers are used to reconstruct the Mel-spectrogram from the reshaped latent vector. Rectified Linear Unit (ReLU) activation functions are employed for the intermediate layers, while a sigmoid activation function is used for the output layer to constrain the output values to the range [0, 1]. The Adam optimizer is used for training.

Convolutional VAE (CVAE). The CVAE architecture employs convolutional and batch normalization layers within the encoder and transposed convolutional and batch normalization layers within the decoder. The encoder consists of five convolutional layers with 512, 256, 128, 64, and

32 filters, kernel sizes of 3 for each layer, and strides of 2, 2, 2, 2, and 2×1 , respectively. Following these layers, dense layers are used to map the extracted features to the latent space, generating the mean μ and log variance $\log \sigma^2$. The latent space dimension is 128.

The decoder mirrors the encoder's structure in terms of layer order but uses transposed convolutions for upsampling. It begins with a dense layer that receives a sample from the latent distribution. A reshape layer then prepares the latent vector for the subsequent transposed convolutional layers. Five transposed convolutional layers are used for upsampling and reconstruction, with filters of 32, 64, 128, 256, and 1, kernel sizes of 3, and strides of 2 for the first four layers. The final transposed convolutional layer has a stride of 2×1 . ReLU activation functions are used for the intermediate layers, and a sigmoid activation function is used for the final upsampling layer to constrain the output to the range $[0, 1]$. The Adam optimizer is employed for training. The input shape for the CVAE is 128×128 . Training uses the standard ELBO objective of Kingma and Welling [50], identical to the MLP-VAE.

Conditional convolutional VAE (CCVAE). The CCVAE [46] extends the CVAE by incorporating conditional information in the form of class labels. This allows for the generation of samples conditioned on specific classes. The encoder takes as input both the Mel-spectrogram and a one-hot encoded label vector. This label vector is concatenated with the flattened downsampled Mel-spectrogram after the feature extraction process. The encoder consists of five convolutional layers (each followed by batch normalization) with 512, 256, 128, 64, and 32 filters, kernel sizes of 3, and strides of 2, 2, 2, 2, and 2×1 , respectively. As with the CVAE, dense layers follow the convolutional layers to produce the mean μ and log variance $\log \sigma^2$ for the latent distribution. The latent space dimension is 128.

The decoder begins with a dense layer that processes the combined latent vector and label condition. This is followed by a reshape operation and then the inverse layers of the encoder are used for upsampling and reconstruction of the Mel-spectrogram. ReLU activations are used for the intermediate layers, and a sigmoid activation is used for the final upsampling layer. The Adam optimizer is used for training. The input shape for the CCVAE is also 128×128 . Training uses the conditional ELBO of Sohn et al. [46], in which the reconstruction term is conditioned on the class label y while the KL divergence term remains unchanged relative to the standard VAE.

β -VAE. The β -VAE [47] modifies the standard VAE objective by introducing a hyperparameter $\beta > 0$ that scales the KL-divergence term of the ELBO, encouraging disentangled latent representations at the cost of weaker reconstruction when $\beta > 1$; the reader is referred to [47] for the formal objective.

The encoder and decoder architectures used in our implementation of the β -VAE are identical to those of the CVAE, employing convolutional and batch normalization layers in the encoder and transposed convolutional and batch normalization layers in the decoder. The only difference with respect to the CVAE is the β -scaled KL term in the loss. Based on empirical evaluation and considering the trade-off between reconstruction quality and disentanglement for our cough sound data, we chose $\beta = 4$ for our experiments.

The latent space dimension is 128, and the input shape is 128×128 . ReLU activation is used for the hidden layers, with a sigmoid activation for the output layer to constrain the output to the range $[0, 1]$. The Adam optimizer is used for training.

Vector quantized VAE (VQ-VAE). The Vector Quantized VAE [48] replaces the continuous latent space of the standard VAE with a discrete codebook of learnable embeddings; each encoder output is mapped to its nearest codebook entry by Euclidean distance, and the decoder reconstructs the input from the quantized representation. The training loss combines a reconstruction term, a codebook-update term, and a

commitment term weighted by a scalar β ; we adopt the formulation of van den Oord et al. [48].

In our implementation, the encoder mirrors the CVAE, employing convolutional and batch-normalization layers to downsample the input Mel-spectrogram, and the decoder uses the matching transposed-convolutional structure to reconstruct it from the quantized codes. We set $\beta = 0.25$ for the commitment-loss weight, following the recommendation in [48]. The latent space has a codebook size of $K = 512$ and $D = 64$, meaning there are 512 embeddings in the codebook, each of dimensionality 64. The input Mel-spectrogram has a shape of 128×128 , corresponding to the number of frequency and time bins. ReLU activation is used for the hidden layers, with a sigmoid activation for the output layer to constrain the output to the range $[0, 1]$. The Adam optimizer is used for training.

To model the discrete latent space of the VQ-VAE, we employ a PixelCNN [51] as a prior. The PixelCNN is an autoregressive model that generates discrete latent codes by modeling the conditional distribution of each code given the previous codes. Unlike standard VAEs, which assume a continuous latent space with a simple prior (e.g., Gaussian), the VQ-VAE uses a discrete latent space. This requires a separate model, such as a PixelCNN, to learn the prior distribution over the discrete latent codes. The PixelCNN is necessary because the VQ-VAE itself does not model the dependencies between the discrete latent codes; it only learns to map inputs to the nearest codebook embeddings. By training a PixelCNN on the discrete latent codes, we can generate new codes that follow the learned prior, enabling the VQ-VAE to produce diverse and realistic samples.

Our PixelCNN architecture begins with an input layer that takes the discrete latent codes from the VQ-VAE as input. The input shape is 16×16 , corresponding to the spatial dimensions of the latent space for a downsampled Mel-spectrogram of shape 128×128 , and the depth is equal to the codebook size $K = 512$. The first layer is a masked convolutional layer with a Type A mask, 128 filters, a kernel size of 7×7 , and a stride of 1. This is followed by 12 additional masked convolutional layers with Type B masks, each with 128 filters, a kernel size of 3×3 , and a stride of 1. Each masked convolutional layer is followed by a gated activation unit, which applies a tanh activation to the filter gate and a sigmoid activation to the gating mechanism.

The architecture includes six residual blocks to improve gradient flow and model capacity. Each residual block consists of a masked convolutional layer with 128 filters, a kernel size of 3×3 , and a stride of 1, followed by a 1×1 convolutional layer to match the number of channels. Skip connections are used to add the input of each block to its output. The final layer is a convolutional layer with 512 filters (equal to the codebook size $K = 512$), a kernel size of 1×1 , and a stride of 1. This layer uses a softmax activation to predict the probability distribution over the codebook indices.

The PixelCNN is trained using the Adam optimizer with a learning rate of 3×10^{-4} and a batch size of 32. Training is performed to maximize the likelihood of the discrete latent codes produced by the VQ-VAE. Once trained, the PixelCNN is used to sample new latent codes, which are then passed through the VQ-VAE decoder to generate new samples.

5.2.3. Inverse audio processing pipeline

To reconstruct time-domain audio from the generated Mel-spectrograms, an inverse processing pipeline is employed. Initially, the normalized spectrograms are rescaled back to their original amplitude range of $[-80, 0]$ dB. These are then converted from the decibel scale to the power scale to approximate the original energy distribution. Because Mel-spectrograms do not retain phase information, the corresponding linear-scale spectrogram is estimated by inverting the Mel filterbank projection. To reconstruct the waveform, the Griffin-Lim algorithm [52] is applied with 64 iterations to iteratively estimate the phase and recover a plausible time-domain signal.

5.2.4. Evaluation of the generated audio samples

Following the conversion of generated Mel-spectrograms back into the time domain, the resulting audio signals undergo post-processing to improve perceptual and statistical quality. Specifically, a band-pass Butterworth filter with a frequency range of 50 Hz to 6000 Hz is applied to attenuate low-frequency background noise and high-frequency artifacts introduced during decoding. Subsequently, an audio-normalization step rescales the waveforms to the $[-1, 1]$ range, ensuring consistency across all generated samples.

To evaluate the fidelity and practical utility of the generated cough sounds, we employ a set of perceptual, statistical, and task-specific metrics following prior work in medical data synthesis and generative modeling, including Kontogiannis et al. [53] and other relevant studies [54–56].

Fréchet audio distance (FAD). FAD [49] is a perceptual metric that compares the distribution of embeddings between real and generated audio data. The embeddings are extracted using a pretrained VGGish model. Let $\mathcal{N}(\mu_r, \Sigma_r)$ and $\mathcal{N}(\mu_s, \Sigma_s)$ be multivariate Gaussians fitted to the embeddings of real and synthetic data, respectively. The FAD is computed as:

$$\text{FAD} = \|\mu_r - \mu_s\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_s - 2(\Sigma_r \Sigma_s)^{1/2}) \quad (15)$$

Lower FAD values indicate that the distribution of synthetic audio closely matches that of real audio, implying higher perceptual realism. This metric is robust to noise and does not require exact correspondence between samples.

Cross-correlation. Cross-correlation measures the similarity of temporal structures between two signals. Given a real signal $x(t)$ and a generated signal $y(t)$ of equal length T , the normalized cross-correlation is defined as:

$$\rho = \frac{1}{T} \sum_{t=1}^T \frac{(x(t) - \bar{x})(y(t) - \bar{y})}{\sigma_x \sigma_y} \quad (16)$$

where $\bar{x}$ and $\bar{y}$ are the means, and σ_x, σ_y are the standard deviations of the respective signals. We compute the correlation using the “valid” mode to minimize edge effects. Higher values of ρ indicate stronger temporal similarity and phase alignment between real and generated cough signals.

Mel cepstral distortion (MCD). MCD [57] quantifies the spectral difference between two audio signals using their Mel-frequency cepstral coefficients (MFCCs). Let $c_t = [c_1^{(t)}, c_2^{(t)}, \dots, c_M^{(t)}]$ and $\hat{c}_t$ be the MFCC vectors for real and synthetic signals at time t , respectively. The MCD is given by:

$$\text{MCD[dB]} = \frac{10}{\ln 10} \sqrt{2 \sum_{m=1}^M (c_m^{(t)} - \hat{c}_m^{(t)})^2} \quad (17)$$

Before computing MCD, we apply Dynamic Time Warping (DTW) to optimally align the sequences. Lower MCD values indicate higher spectral similarity between the signals.

Train on real, test on synthetic (TRTS). The TRTS evaluation protocol [55,56] assesses the realism of the generated audio by measuring how well a classifier trained solely on real data performs when tested on synthetic samples. High classification accuracy in this setting suggests that the synthetic samples are statistically and semantically aligned with the true data distribution, preserving key disease-related characteristics. For this purpose, we employ a multi-class SVM classifier trained on real cough samples and evaluated on the corresponding synthetic samples.

Train on synthetic, test on real (TSTR). TSTR [55,56] is designed to evaluate the diversity and generalization capacity of synthetic data. In this setup, a multi-class SVM classifier is trained exclusively on generated samples and tested on real data. Strong performance under TSTR conditions indicates that the synthetic dataset captures a broad and representative distribution of real cough events. In contrast, poor performance may suggest issues such as mode collapse or insufficient variability in the generated samples.

5.3. Experiments and evaluation

To assess the effectiveness of the proposed VAE augmentation module, we generated 1000 synthetic audio samples per class for each VAE architecture and randomly selected 200 samples for quality assessment.

Variational Autoencoders support two primary strategies for sample generation: (i) *reconstruction-based generation*, where real inputs are encoded into the latent space and decoded back into audio, and (ii) *sampling-based generation*, where latent vectors are drawn from the learned prior distribution and decoded into novel samples. Although reconstruction-based generation typically yields higher-fidelity outputs, it risks data leakage and is inherently limited in diversity by the number of available test samples. For these reasons, we adopt the sampling-based strategy, which provides greater variability while ensuring no reconstruction of test-set samples.

All real cough events in the test set were used as reference data against which synthetic samples were evaluated. Three widely used metrics were computed: Fréchet Audio Distance (FAD) [49], Cross-Correlation similarity, and Mel Cepstral Distortion (MCD) [57]. These metrics capture perceptual fidelity and spectral-temporal similarity between real and synthetic coughs.

To assess the *practical utility* of synthetic samples, we conducted classification-based experiments using Train-on-Real-Test-on-Synthetic (TRTS), Train-on-Synthetic-Test-on-Real (TSTR), and Hybrid (Real + Synthetic) training. SVM classifiers were used with MFCC features, following the findings of Section 4. Finally, a subset of generated samples was incorporated into the training set of the Disease Analysis Module to balance minority disease classes, and the resulting improvements in end-to-end performance were quantified through the Voting Mechanism.

5.3.1. Generative fidelity and cross-domain performance of VAE models

In Table 7, we report the generative fidelity metrics (Fréchet Audio Distance, Cross-Correlation similarity, and Mel Cepstral Distortion) for each VAE architecture across the three target classes (Asthma, COVID-19, and Healthy). The same table also reports TRTS and TSTR classification performance. These results quantify both the acoustic realism of the generated cough samples and their practical utility for cross-domain generalization.

A clear hierarchy emerges across the architectures. The *MLP-VAE* consistently produces the lowest-quality samples, with the highest FAD values (> 2.6), lowest cross-correlation (0.27–0.30), and the largest MCD values (23–25). Correspondingly, it achieves the weakest classification performance, with TRTS scores around 52–53% and TSTR scores around 50–51%. These results confirm that fully connected architectures fail to capture the structured temporal and spectral patterns characteristic of cough acoustics.

In contrast, all *convolution-based VAEs* substantially improve generative fidelity. The *CVAE* reduces MCD to approximately 5.55, increases cross-correlation above 0.60, and achieves TRTS/TSTR values near 74% and 70%, indicating that its samples capture disease-consistent spectral envelopes. The *CCVAE* further improves fidelity with MCD values around 5.48 and cross-correlation up to 0.63, resulting in TSTR scores exceeding 72%.

The β -VAE continues this trend, achieving lower distortion (MCD ≈ 5.46) and higher similarity (CC up to 0.64), with TRTS/TSTR performance rising to roughly 76% and 73%, respectively.

Table 7

Generative fidelity and cross-domain generalization performance for each VAE architecture. Fidelity metrics (FAD, CC, MCD) are computed between real cough samples from the test set and 200 randomly selected synthetic samples generated per model. Lower FAD and MCD, and higher cross-correlation (CC), indicate higher acoustic realism. TRTS and TSTR report SVM classification performance under real→synthetic and synthetic→real domain shifts, respectively (mean $\pm$ std across 5 folds).

Model	Class	FAD ($\downarrow$)	CC ($\uparrow$)	MCD ($\downarrow$)	TRTS F1 (%)	TSTR F1 (%)
MLP-VAE	Asthma	2.90 $\pm$ 0.03	0.29 $\pm$ 0.06	25.1 $\pm$ 1.4	52.3 $\pm$ 2.2	50.9 $\pm$ 2.3
	COVID-19	2.63 $\pm$ 0.02	0.27 $\pm$ 0.07	23.6 $\pm$ 1.3	51.8 $\pm$ 2.0	50.4 $\pm$ 2.1
	Healthy	2.87 $\pm$ 0.03	0.30 $\pm$ 0.07	24.9 $\pm$ 1.5	53.0 $\pm$ 2.1	51.1 $\pm$ 2.4
CVAE	Asthma	0.44 $\pm$ 0.02	0.60 $\pm$ 0.04	5.55 $\pm$ 0.22	73.9 $\pm$ 1.7	70.3 $\pm$ 1.9
	COVID-19	0.48 $\pm$ 0.03	0.58 $\pm$ 0.04	5.63 $\pm$ 0.25	73.1 $\pm$ 1.6	69.8 $\pm$ 1.8
	Healthy	0.46 $\pm$ 0.02	0.62 $\pm$ 0.03	5.58 $\pm$ 0.23	74.1 $\pm$ 1.6	69.9 $\pm$ 1.7
CCVAE	Asthma	0.41 $\pm$ 0.02	0.62 $\pm$ 0.03	5.48 $\pm$ 0.19	75.0 $\pm$ 1.6	72.2 $\pm$ 1.7
	COVID-19	0.45 $\pm$ 0.03	0.60 $\pm$ 0.04	5.56 $\pm$ 0.22	74.6 $\pm$ 1.7	71.4 $\pm$ 1.7
	Healthy	0.42 $\pm$ 0.02	0.63 $\pm$ 0.02	5.52 $\pm$ 0.21	75.3 $\pm$ 1.5	72.0 $\pm$ 1.5
β -VAE	Asthma	0.40 $\pm$ 0.02	0.63 $\pm$ 0.03	5.46 $\pm$ 0.18	75.7 $\pm$ 1.5	73.0 $\pm$ 1.6
	COVID-19	0.44 $\pm$ 0.03	0.62 $\pm$ 0.03	5.53 $\pm$ 0.20	75.2 $\pm$ 1.6	72.6 $\pm$ 1.6
	Healthy	0.41 $\pm$ 0.02	0.64 $\pm$ 0.02	5.49 $\pm$ 0.17	75.8 $\pm$ 1.5	73.1 $\pm$ 1.7
VQ-VAE	Asthma	0.38 $\pm$ 0.02	0.64 $\pm$ 0.02	5.40 $\pm$ 0.15	77.0 $\pm$ 1.4	75.1 $\pm$ 1.5
	COVID-19	0.43 $\pm$ 0.02	0.63 $\pm$ 0.02	5.50 $\pm$ 0.18	76.4 $\pm$ 1.4	74.6 $\pm$ 1.5
	Healthy	0.41 $\pm$ 0.02	0.65 $\pm$ 0.02	5.45 $\pm$ 0.16	77.1 $\pm$ 1.3	75.3 $\pm$ 1.4

Table 8

Performance of the disease analysis Module (DAM) after retraining with class-balanced VAE-based augmentation. The Voting Mechanism outputs final 3-class predictions (Asthma, COVID-19, Healthy) using Bayesian Posterior Fusion. Metrics reflect mean $\pm$ std across 5-fold speaker-independent cross-validation. Δ F1 denotes the absolute F1-score improvement relative to the DSP-Augmented Baseline. The **Synth./fold** column reports the mean number of synthetic cough segments drawn per fold from the per-class 16,000-sample generative pool of Table 6 at the principal 3.5 $\times$ per-class augmentation ratio; values are architecture-invariant across the five VAE variants compared here. The 3.5 $\times$ operating point is selected empirically from the magnitude-sensitivity sweep of Table 11.

Model	Class	Synth./fold	Specificity (%)	Recall (%)	F1-score (%)	Δ F1 (%)
Baseline (DSP)	Asthma	0	73.2 $\pm$ 2.6	71.0 $\pm$ 2.4	72.1 $\pm$ 2.3	–
	COVID-19	0	71.9 $\pm$ 2.5	69.8 $\pm$ 2.4	70.8 $\pm$ 2.3	–
	Healthy	0	81.4 $\pm$ 1.8	84.2 $\pm$ 1.6	82.7 $\pm$ 1.7	–
MLP-VAE	Asthma	1291	72.4 $\pm$ 2.5	70.1 $\pm$ 2.5	71.2 $\pm$ 2.4	–0.9
	COVID-19	1524	71.1 $\pm$ 2.4	68.9 $\pm$ 2.5	70.0 $\pm$ 2.4	–0.8
	Healthy	1197	80.6 $\pm$ 1.9	83.5 $\pm$ 1.7	82.0 $\pm$ 1.8	–0.7
CVAE	Asthma	1291	75.6 $\pm$ 2.3	73.4 $\pm$ 2.1	74.6 $\pm$ 2.1	+2.5
	COVID-19	1524	74.3 $\pm$ 2.2	72.1 $\pm$ 2.1	73.3 $\pm$ 2.1	+2.5
	Healthy	1197	83.5 $\pm$ 1.6	86.3 $\pm$ 1.4	84.7 $\pm$ 1.5	+2.0
CCVAE	Asthma	1291	76.5 $\pm$ 2.1	74.3 $\pm$ 2.0	75.5 $\pm$ 2.0	+3.4
	COVID-19	1524	75.4 $\pm$ 2.1	73.2 $\pm$ 2.0	74.4 $\pm$ 2.0	+3.6
	Healthy	1197	84.3 $\pm$ 1.5	87.1 $\pm$ 1.4	85.5 $\pm$ 1.4	+2.8
β -VAE	Asthma	1291	77.8 $\pm$ 2.0	75.6 $\pm$ 1.8	76.7 $\pm$ 1.9	+4.6
	COVID-19	1524	76.6 $\pm$ 1.9	74.5 $\pm$ 1.9	75.6 $\pm$ 1.9	+4.8
	Healthy	1197	85.6 $\pm$ 1.4	88.2 $\pm$ 1.3	86.7 $\pm$ 1.3	+4.0
VQ-VAE	Asthma	1291	79.2 $\pm$ 1.8	77.1 $\pm$ 1.7	78.2 $\pm$ 1.7	+6.1
	COVID-19	1524	78.0 $\pm$ 1.8	76.0 $\pm$ 1.7	77.0 $\pm$ 1.7	+6.2
	Healthy	1197	87.0 $\pm$ 1.3	89.6 $\pm$ 1.2	88.2 $\pm$ 1.2	+5.5

The VQ-VAE delivers the highest fidelity among all models. It achieves the lowest FAD values (as low as 0.38), the highest cross-correlation scores (0.64–0.65), and the lowest MCD values (5.40–5.50). Importantly, it also yields the strongest cross-domain classification results, with TRTS values around 77% and TSTR values exceeding 75%.

Taken together, these results show that *generative augmentation is only beneficial when synthetic samples exhibit high acoustic fidelity and preserve disease-specific structure*. Convolutional and codebook-based VAEs satisfy this requirement, whereas the MLP-VAE does not, producing low-fidelity samples that lack diagnostic value.

Within every cross-validation fold, all three classes (asthma, COVID-19, and healthy) were augmented from the per-class 16,000-sample synthetic pool of the corresponding VAE variant at the principal 3.5 $\times$

per-class ratio, defined as the augmented-to-original training-window ratio applied independently to each class so that the resulting per-class size becomes $3.5 \times N_{\text{orig}}$. The choice of 3.5 $\times$ as the operating point is empirically motivated by the magnitude-sensitivity sweep of Section 5.3.3 (Table 11), which shows that performance peaks at this ratio and degrades on either side. Draws were independent across folds and disjoint by construction, so no synthetic segment appeared in more than one training split within a single run. The **Synth./fold** column of Table 8 reports the resulting mean per-fold draw count; it is architecture-invariant (the VAE variant determines *which* pool is sampled, not *how many* samples are drawn).

Table 9 reports the corresponding 3-class confusion matrix for this best configuration, computed on the 20% held-out test partition

Table 9

Three-class confusion matrix of the best-performing disease analysis Module (DAM) configuration: VQ-VAE synthetic augmentation with Bayesian Posterior Fusion, evaluated on the 20% held-out test partition of Table 3 ($N = 2006$ cough segments; Asthma: 645, COVID-19: 762, Healthy: 599). Rows correspond to ground-truth labels, columns to predicted labels.

True	Predicted			Total
	Asthma	COVID-19	Healthy	
Asthma	504	116	25	645
COVID-19	129	587	46	762
Healthy	12	59	528	599
Total	645	762	599	2006

Table 10

Coverage–F1 trade-off of the abstain rule (Eq. (14)) after VAE-based augmentation, measured on the held-out validation folds of the three-class Voting Mechanism under the VQ-VAE + Bayesian Posterior Fusion configuration (best row of Table 8). Columns mirror Table 5; the rightmost column reports $\Delta F1$ against the DSP-augmented operating point at the same τ . The default operating point $\tau^* = 0.80$ is highlighted in bold.

τ	Coverage (%)	Covered F1 (%)	$\Delta F1$ vs. Table 5 (pp)
0.50	99.6	81.2	+5.9
0.60	95.8	84.5	+5.4
0.70	88.1	87.6	+4.4
0.75	81.4	89.9	+4.1
0.80*	74.2	91.7	+3.3
0.85	63.6	93.4	+2.5
0.90	50.1	95.2	+1.8

(Table 3). The healthy class concentrates the least off-diagonal mass (71 of 599 healthy segments misclassified, 11.9%), whereas the two disease classes exhibit the largest inter-class confusion between themselves. Confusion into the healthy class (i.e., false negatives for the disease-positive classes) is comparatively rare, which is the operationally preferable failure mode for a first-line screening aid that is followed by the abstain rule of Section 4.4.1.

To quantify how VAE-based augmentation interacts with the screening safeguard itself, we repeated the coverage–F1 sweep of Table 5 under the best-performing configuration of Table 8, i.e., VQ-VAE synthetic augmentation combined with Bayesian Posterior Fusion. Table 10 reports the resulting operating points over the same τ grid. Covered-set F1 improves monotonically over the DSP-only baseline at every threshold, and coverage simultaneously increases. At the default operating point $\tau^* = 0.80$ the DAM abstains on only 25.8% of inputs (versus 31.5% in Table 5) and attains 91.7% covered-set F1 (versus 88.4%), a +3.3 pp gain on covered inputs with a simultaneous +5.7 pp coverage gain. Gains narrow as τ grows (covered-set accuracy saturates near the per-class ceiling established by Table 8) but remain strictly positive across the entire sweep. VAE augmentation therefore improves the DAM on two distinct axes: it raises the unfiltered three-class F1 of Table 8 and, independently, tightens the full coverage–F1 envelope of the deployed abstain rule reported in Table 10.

5.3.2. Impact of VAE-based augmentation on disease analysis module performance

The Baseline (DSP) setup with Bayesian Posterior Fusion yields F1-scores of 72.1% (Asthma), 70.8% (COVID-19), and 82.7% (Healthy). Consistent with the generative fidelity results in Table 7, the MLP-VAE adversely affects performance. All three classes experience a small decrease (–0.7% to –0.9%), indicating that lower-quality synthetic samples introduce ambiguous or noisy decision boundaries. This degradation confirms that sample fidelity is essential when augmenting classifiers operating in low-data regimes.

In contrast, all convolution-based VAEs improve DAM performance. The CVAE and CCVAE architectures provide moderate but consistent gains (+2–3.6% absolute F1), suggesting that their synthetic samples effectively enrich intra-class variability without distorting class structure. The improvements are reflected in both specificity and recall, indicating reduced false positives and more stable disease detection.

The β -VAE further strengthens performance, achieving +4.0–4.8% F1 improvements across classes. The highest gains arise from the VQ-VAE, which increases F1-scores by +6.1% (Asthma), +6.2% (COVID-19), and +5.5% (Healthy). These improvements are also stable across specificity and recall, indicating both better class discrimination and more reliable rejection of low-confidence predictions.

Importantly, because the Voting Mechanism employs Bayesian Posterior Fusion, these improvements occur under a decision rule that adaptively weights classifier outputs based on cross-window consistency. The mechanism assigns higher influence to classifiers exhibiting stable predictions across temporal windows, and conservatively favors the Healthy class when both classifiers are uncertain. Thus, the improved F1-scores reflect not only better classifier generalization but also better *certainty* in disease predictions. The VQ-VAE's gains under such adaptive weighting suggest that it generates diagnostically faithful cough patterns that meaningfully expand the training distribution.

The VQ-VAE's F1 dominance documented above is reinforced orthogonally by the abstain-rule operating-point sweep of Table 10: under the same VQ-VAE + Bayesian Posterior Fusion configuration, the coverage–F1 envelope dominates the DSP-augmented envelope of Table 5 at every threshold $\tau \in [0.50, 0.90]$, with +3.3 pp covered-set F1 and +5.7 pp coverage at the default $\tau^* = 0.80$ (91.7% covered F1 at 74.2% coverage, versus 88.4% at 68.5%). The VQ-VAE's segment-level ranking therefore propagates to the deployed screening safeguard of Section 4.4.1 without any retraining or threshold retuning, confirming that the ranking is not a threshold-specific artefact of the unfiltered Voting-Mechanism evaluation.

Taken together, these findings demonstrate that *high-fidelity generative augmentation substantially strengthens disease classification*. Conversely, low-fidelity augmentation (MLP-VAE) can degrade performance, underscoring the necessity of architectural inductive biases when modeling bioacoustic signals.

5.3.3. Sensitivity to augmentation magnitude

The principal configuration of Table 8 adopts a 3.5 $\times$ per-class augmentation ratio, but this choice itself is an empirical question: insufficient augmentation leaves residual class imbalance and limited acoustic-mode coverage, whereas excessive augmentation lets synthetic-segment artefacts dominate the SVM decision boundary at the expense of real-data evidence. To characterise this trade-off, we sweep the per-class augmentation ratio $r = (N_{\text{orig}} + N_{\text{synth}})/N_{\text{orig}}$ on a uniform 0.5-step grid from $r = 1.0$ (DSP-only baseline) to $r = 6.0$. Synthetic samples are redrawn within each fold from the per-class 16,000-sample VQ-VAE pool of Table 6 (the VQ-VAE generator is selected as the dominant variant of Table 7); the pool is sized so that even the most extreme $r = 6.0$ row (≈ 3000 synthetic windows per fold for COVID-19) consumes only 5-fold disjoint draws from a single generative population. All three classes are augmented at the same ratio r , so the relative class proportions are preserved while the absolute training-window count scales linearly. Each row of Table 11 is obtained by retraining DAM under the speaker-independent 5-fold cross-validation protocol of Section 4.5, with disjoint per-fold draws across folds.

The sweep reveals a clear inverted-U relationship between augmentation magnitude and downstream classification F1, with the peak located at $r = 3.5$ across all three classes and three distinct regimes on either side. In the *pre-peak* regime ($r \in [1.0, 3.5]$), gains scale near-monotonically with r as additional synthetic windows fill out the latent acoustic-mode coverage of each class: mean $\Delta F1$ rises smoothly from +1.5 pp at $r = 1.5$ through +4.3 pp at $r = 2.5$ and +5.4 pp at $r = 3.0$ to the peak +5.9 pp at $r = 3.5$, where the per-fold

Table 11

Sensitivity of disease-classification performance to VAE-augmentation magnitude (VQ-VAE generator with Bayesian Posterior Fusion). The per-class augmentation ratio $r = (N_{\text{orig}} + N_{\text{synth}})/N_{\text{orig}}$ is applied independently to all three classes; $r = 1.0$ is the DSP-only baseline. Synth./fold reports the mean per-fold synthetic-sample draw count for each class. Mean $\Delta F1$ is the across-class average F1 improvement relative to the $r = 1.0$ baseline. Grid spacing is uniform at 0.5 from $r = 1.0$ to $r = 6.0$; the $r = 3.5$ configuration (bold) is the principal operating point used throughout Section 5 and reproduced as the VQ-VAE row of Table 8.

Ratio	Synth./fold			F1-score (%)			Mean $\Delta F1$ (pp)
	Asthma	COVID-19	Healthy	Asthma	COVID-19	Healthy	
1.0×	0	0	0	72.1 ± 2.3	70.8 ± 2.3	82.7 ± 1.7	–
1.5×	258	305	239	73.7 ± 2.1	72.4 ± 2.1	84.0 ± 1.5	+1.5
2.0×	516	609	479	75.2 ± 1.9	74.0 ± 1.9	85.4 ± 1.4	+3.0
2.5×	775	914	718	76.5 ± 1.8	75.3 ± 1.8	86.7 ± 1.3	+4.3
3.0×	1033	1219	958	77.7 ± 1.7	76.5 ± 1.7	87.7 ± 1.2	+5.4
3.5×	1291	1524	1197	78.2 ± 1.7	77.0 ± 1.7	88.2 ± 1.2	+5.9
4.0×	1549	1828	1436	77.3 ± 1.8	76.1 ± 1.8	87.3 ± 1.3	+5.0
4.5×	1807	2133	1675	75.7 ± 2.0	74.5 ± 2.0	85.7 ± 1.5	+3.4
5.0×	2066	2438	1915	73.7 ± 2.3	72.5 ± 2.2	83.7 ± 1.7	+1.4
5.5×	2324	2742	2155	71.4 ± 2.7	70.2 ± 2.6	81.3 ± 2.0	−0.9
6.0×	2582	3047	2394	68.9 ± 3.0	67.7 ± 3.0	78.8 ± 2.4	−3.4

synthetic draw count (≈ 1300 for Asthma, ≈ 1500 for COVID-19, ≈ 1200 for Healthy) is approximately 60% of the per-fold real training set. In the *post-peak above-baseline* regime ($r \in [3.5, 5.0]$), performance degrades gradually: mean $\Delta F1$ falls to + 5.0 pp at $r = 4.0$, + 3.4 pp at $r = 4.5$, and + 1.4 pp at $r = 5.0$, indicating that the additional synthetic windows beyond the peak contribute progressively less novel acoustic variability and progressively more redundancy. From $r = 5.5$ onward the curve enters a *strict-below-baseline* regime in which augmentation actively harms classification: mean $\Delta F1$ falls to −0.9 pp at $r = 5.5$ and reaches −3.4 pp at $r = 6.0$ (Asthma 68.9, COVID-19 67.7, Healthy 78.8), with standard deviation nearly doubling relative to the peak (Asthma: 1.7 → 3.0; COVID-19: 1.7 → 3.0; Healthy: 1.2 → 2.4).

This standard deviation expansion signals that the 5-fold draws progressively populate fold-specific generator-bias regions of the codebook rather than novel respiratory-acoustic content, so the SVM decision boundary drifts toward synthetic-segment artifacts at the expense of real-data evidence. The 3.5× ratio is therefore adopted as the principled operating point throughout the manuscript, balancing the class-balance and acoustic-coverage benefits of generative augmentation against the generator-induced distributional drift that emerges once synthetic windows substantially outnumber real ones.

5.4. End-to-end pipeline evaluation

The results reported in Sections 3 and Section 4.5 evaluate the Cough Detection Module and the Disease Analysis Module in isolation: the CDM is scored on window-level and event-level cough-vs.-non-cough detection (Table 2), while the DAM is scored on pre-segmented cough events drawn directly from the Kuluozturk corpus (Tables 4, Tables 8 and 9). To characterize the fully deployed cascade, we additionally evaluate the end-to-end pipeline on the same 20% held-out Kuluozturk recordings used to construct Table 9, totaling 477 recordings (157 Asthma, 181 COVID-19, 139 Healthy), selected under the speaker-independent 80/20 split of Section 4.2 so that no subject appears in both training and evaluation.

Each recording is processed from raw audio through the three cascaded stages detailed in Section 6: 300 ms windowing and 1D-CNN scoring (Section 3), event detection by EDEM (Section 3), and disease classification by the Bayesian Posterior Fusion of the two SVMs under the best-performing VQ-VAE augmentation configuration (Section 5) with the abstain threshold $\tau^* = 0.80$ defined in Eq. (14). Recording-level predictions are obtained by a majority vote over the per-event posterior-fusion decisions of the DAM; recordings in which the CDM detects no cough event, or in which every detected event triggers the abstain rule, are labeled as *Uncertain*.

Table 12

End-to-end pipeline evaluation: 3-class-plus-Uncertain confusion matrix at the recording level on the 20% held-out Kuluozturk split ($N = 477$; 157 Asthma, 181 COVID-19, 139 Healthy). The pipeline runs raw audio through CDM detection, EDEM event extraction, and DAM classification under the best-performing VQ-VAE + Bayesian Posterior Fusion configuration with the abstain threshold $\tau^* = 0.80$ per Eq. (14); recording-level predictions are produced by majority vote over per-event DAM decisions. The *Uncertain* column aggregates two failure modes: 12 recordings in which the CDM detected no cough event (CDM-miss) and 4 recordings in which every detected cough event was rejected by the abstain rule at $\tau^* = 0.80$ (abstain-driven), totaling 16 recordings (3.4% of the split).

True \ Predicted	Asthma	COVID-19	Healthy	Uncertain	Total
Asthma	121	21	10	5	157
COVID-19	23	139	12	7	181
Healthy	12	10	113	4	139
Total	156	170	135	16	477

Cascade performance analysis on the held-out split: the CDM achieves a per-event Detection Ratio of 86.2% on the Kuluozturk recordings, representing a decrease from 90.0% on the pilot set of Table 2, consistent with the different acquisition chains between the two corpora. Because most recordings contain multiple cough events, the *recording-level* cough-capture rate is 97.5%, with 12/477 recordings (2.5%) yielding no detected cough events and falling back to *Uncertain* on CDM grounds. The CDM false-positive rate is approximately 3.8% of detected events. A further 4/477 recordings (0.8%) had every detected cough event rejected by the abstain rule at $\tau^* = 0.80$ and therefore also default to *Uncertain*, for a combined total of 16/477 *Uncertain* recordings (3.4% of the split). Table 12 reports the resulting 3-class-plus-Uncertain confusion matrix.

Per-class specificity, recall, and F1-score derived from Table 12 are 77.6%/77.1%/77.3% (Asthma), 81.8%/76.8%/79.2% (COVID-19), and 83.7%/81.3%/82.5% (Healthy), yielding a macro-averaged recording-level F1 of 79.7% and an overall accuracy of 78.2% (counting *Uncertain* as an incorrect prediction). Compared to the segment-level DAM-only macro-F1 of 81.1% reported for the VQ-VAE configuration in Table 8, which represents the strongest segment-level configuration in the present study, the end-to-end recording-level metric drops by 1.4 percentage points. Restricting the analysis to covered recordings only, i.e., excluding the 16 *Uncertain* recordings comprising 12 CDM-misses and 4 abstain-driven cases, the covered-set macro-F1 is 81.0%, which matches the DAM-only segment-level performance to within rounding. This decomposition confirms that the end-to-end degradation is attributable almost entirely to CDM misses and abstain-rule rejections

on recordings with sparse or acoustically atypical coughs. Conditional on successful cough detection and a non-abstaining DAM output, majority-vote aggregation at the recording level reproduces the segment-level DAM performance.

Although this end-to-end evaluation utilizes a retrospective corpus rather than prospective recordings from unconstrained environments, it serves as a robust surrogate for assessing system performance in a real-world deployment scenario. Validating these findings through a prospective field study on a heterogeneous clinical cohort remains a primary objective for future research, as further detailed in Section 7.

6. Smartphone deployment

The proposed trained framework was deployed as a native Android application that executes the *entire data processing pipeline on the device*, from continuous audio acquisition to final disease prediction. The application integrates both the Cough Detection Module and the Disease Analysis Module, without requiring network connectivity at any stage. This design ensures strict user privacy, low operational costs, and reliable operation in resource-constrained environments.

6.1. Model packaging and runtime environment

For the CDM, the 1D-CNN cough classifier was converted and optimized to TensorFlow Lite, which serves as the on-device inference engine for window-level cough versus non-cough classification. The DAM relies on two SVM models and their associated preprocessing pipelines. These components were implemented using `crossai-ts` [58] built on top of `scikit-learn` [59] and serialized as `joblib` objects. The complete Android application package occupies approximately 200 MB, whereas the cough analysis pipeline itself, including the TFLite model, preprocessing pipelines, and the two SVM classifiers, requires only 400 kB.

6.2. On-device audio processing pipeline

Audio is continuously recorded at 16 kHz using the smartphone microphone. The signal is buffered in 10-second segments and processed sequentially by the CDM. Each buffer is segmented into fixed-length analysis windows of 300 ms (4800 samples). For the CDM, a 75% overlap is applied, resulting in a hop size of 75 ms and approximately 133 windows per 10-second buffer. The DAM operates on non-overlapping windows, which yield approximately 33 windows per 10-second buffer.

For each window, the TFLite model outputs cough and non-cough probabilities, which are passed to the Event Detection and Extraction Mechanism (EDEM). The EDEM performs spline interpolation, temporal smoothing, and duration-based thresholding to identify continuous cough episodes. These episodes are then passed to the DAM.

Within the DAM, the corresponding preprocessing steps, including filtering, feature extraction, and scaling, are executed on the device. The two SVM classifiers output probabilities for asthma and COVID-19. Episodes shorter than 300 ms are not analysed. Finally, the Voting Mechanism aggregates the outputs and produces a single three-class decision for each episode.

6.3. Runtime performance

The application processes a 10-second audio buffer in less than 4.3 s on a mid-range Android smartphone. This corresponds to a real-time factor of 0.43, meaning that 0.43 s of computation is required for each 1 s of audio. At the window level, this corresponds to approximately 32 milliseconds of processing per 300 ms CDM window when averaging across all 133 windows.

The system operates faster than real-time while executing both the CDM and DAM pipelines entirely on the smartphone. By avoiding off-device computation, the framework supports continuous, privacy-preserving respiratory monitoring suitable for real-world deployment.

Table 13

Per-component inference-time computational complexity of the deployed on-device pipeline. MAC counts are reported per 300 ms analysis window. Memory figures correspond to the serialized artefact as bundled in the Android application package referenced in Section 6; the aggregate 400 kB total is consistent with the pipeline size quoted in Section 6.

Component	Params/SVs	MACs/window	Memory
1D-CNN (CDM)	86,882 params	177.5 M	90 kB (int8 TFLite)
Asthma SVM (DAM)	1542 SVs $\times$ 20 feat	30.8 k	128 kB
COVID-19 SVM (DAM)	1806 SVs $\times$ 19 feat	34.3 k	142 kB
Preprocessing + scalars	–	0.6 M	40 kB
Pipeline total	–	–	400 kB

6.4. Computational complexity

Complementing the wall-clock measurement of Section 6, we characterize the inference-time computational complexity of the deployed pipeline on a per-module and per-window basis. The VAE Augmentation Module of Section 5 is train-time only, is not serialized into the Android application package, and is therefore excluded from the on-device budget. Only the CDM 1D-CNN, the two DAM SVMs, and their shared preprocessing stages contribute to the inference cost.

Table 13 summarizes, for each component, the parameter count (or number of support vectors $\times$ feature dimensionality), the multiply-accumulate operations (MACs) per 300 ms analysis window, and the on-device memory footprint. The 1D-CNN is reported in its TensorFlow Lite int8-quantized form as deployed, while the two SVMs are reported as serialized `joblib` objects under float32 storage of the support-vector set, dual coefficients, and intercepts. The preprocessing budget aggregates the IIR filter coefficients, MFCC/PSD feature extractors, and scaler parameters shared across the two DAM pipelines.

Expressed asymptotically per 300 ms window, the 1D-CNN cost is $\mathcal{O}\left(\sum_{\ell} L_{\ell} \cdot C_{\ell}^{\text{in}} \cdot C_{\ell}^{\text{out}} \cdot k_{\ell}\right)$ with L_{ℓ} being the layer- ℓ output length under valid padding (successively 4798, 4796, 1598 after the stride-3 max-pool, 1596, and 1594), $C_{\ell}^{\text{in}}, C_{\ell}^{\text{out}} \in \{1, 64, 128\}$ the channel counts, and $k_{\ell} = 3$ the kernel size; summing across the four convolutional layers yields the 177.5 M MACs reported in Table 13. Each SVM evaluation is $\mathcal{O}(|\text{SV}| \cdot F)$, with $F = 20$ for the Asthma classifier and $F = 19$ for the COVID-19 classifier. The MFCC/PSD front-end is dominated by the 512-point STFT at a 128-sample hop, which is $\mathcal{O}(N \log N)$ and contributes the 0.6 M-MAC preprocessing term.

At the 10-second-buffer level, aggregating 133 CDM windows assuming a 75 ms hop (≈ 23.6 GMACs) with at most 33 DAM windows (≈ 2.1 MMACs combined across both SVMs) and the auxiliary EDEM spline/smoothing and Voting Mechanism stages (negligible relative to the convolutional cost) yields an end-to-end inference budget consistent with the 4.3 s wall-clock measurement of Section 6. Effective throughput of approximately 5.5 GMACs·s⁻¹ on the mid-range Android smartphone used for benchmarking.

The CDM 1D-CNN therefore dominates the per-buffer compute budget by roughly four orders of magnitude over the DAM SVMs, which is expected given that disease-level inference is performed only on the comparatively rare windows that the CDM has already classified as cough events. This asymmetry also motivates the compact 1D-CNN design choices of Section 3: reducing the CDM convolutional workload has a near-linear effect on pipeline-level latency, whereas reducing SVM complexity does not.

7. Limitations

While the proposed framework establishes a scalable foundation for ubiquitous, on-device respiratory health monitoring, its current scope is defined by specific limitations that provide a necessary roadmap for future clinical integration.

Data-availability gap for smartphone-recorded multi-disease cough corpora. Publicly available cough datasets acquired through commodity smartphone microphones, such as COUGHVID [18], Coswara [27], and Kuluozturk [25], are predominantly labeled for asthma, COVID-19, and healthy controls. While this data scarcity necessitated restricting the current scope to these three categories, it served as the primary motivation for the multi-variant VAE framework developed in this study, which successfully demonstrated the utility of synthetic data in mitigating such gaps. Emerging tuberculosis resources [60–62] indicate promising future directions but currently present restrictive access terms or acquisition-chain differences relative to our commodity-smartphone premise. The broader-coverage strategies enabled by these resources are outlined in the final paragraph of this section.

Deployment scope and comorbidities. The framework is intended as a first-line passive screening aid in settings where asthma and COVID-19 represent the dominant respiratory presentations. Concretely, the envisaged deployment settings include outpatient asthma management, post-COVID respiratory follow-up, and general-practice triage during seasonal viral surges. However, the model has not been validated on subjects with concurrent respiratory comorbidities (e.g., a patient with chronic asthma currently suffering from an acute viral infection), which may introduce confounding acoustic features. The framework is a screening tool and must not be interpreted as a stand-alone differential-diagnostic device.

Validation cohort and acoustic variability. All DAM experiments were conducted on the Kuluozturk corpus, collected at a single clinical site (Firat University Hospital). Cross-site generalization has not been evaluated here, and real-world performance may diverge from the in-distribution results reported in Section 4.5 due to variations in ambient noise, room reverberation, device-to-subject distance, and hardware-specific microphone frequency responses. The gold-standard validation for the real-world deployment claim remains a prospective multi-centre clinical study on a cohort of healthy, asthma, and COVID-19 subjects using the Android application described in Section 6. The retrospective end-to-end analysis on the 20% speaker-independent held-out split reported in Section 5.4 is the closest assessment possible without new prospective data collection.

Path to broader coverage. Two viable paths exist for extending the disease taxonomy without compromising the on-device premise: (i) the release of larger, authorized smartphone-recorded multi-disease datasets, or (ii) the development of domain-adaptation methods capable of bridging high-fidelity clinical-microphone corpora to commodity-smartphone inference. Both remain active research directions and are prerequisites for the expanded disease taxonomy outlined as future work in Section 8.

8. Conclusion

This study establishes a scalable foundation for ubiquitous, on-device respiratory health monitoring. By integrating a high-precision Cough Detection Module with a novel, multi-variant Generative Augmentation Module and a robust Disease Analysis Module, the resulting pipeline transforms raw smartphone audio into reliable clinical insights while maintaining strict user privacy.

The CDM effectively identifies cough events from continuous audio by combining window-level 1D-CNN classification with a temporal Event Extraction and Detection Mechanism. The module achieved a Detection Ratio of 90.0% on the pilot evaluation set (Table 2) and a recording-level cough-capture rate of 97.5% on the held-out Kuluozturk split, with an estimated false-positive rate of ~3.8% as detailed in Section 5.4.

The DAM utilized two parallel SVM classifiers combined through Bayesian Posterior Fusion and trained with VQ-VAE-generated synthetic coughs to address class imbalance. On the 20% speaker-independent

held-out Kuluozturk split, this configuration achieved a segment-level macro-F1 of 81.1%(Table 8), representing a 6% absolute gain over the non-augmented baseline. A complementary abstain-rule layer (Section 4.4.1, $\tau^* = 0.80$) lifted covered-set macro-F1 to 91.7% at 74.2% coverage, providing a lightweight screening safeguard against low-confidence inputs.

A full-cascade end-to-end evaluation on 477 held-out recordings (Section 5.4) yielded a recording-level macro-F1 of 79.7% and an accuracy of 78.2%. The complete pipeline runs on a mid-range Android smartphone within a 400 kB memory budget and requires approximately 4.3 s per 10-second buffer (Section 6.4). No audio or intermediate features are transmitted off the device, confirming the feasibility of privacy-preserving, low-cost respiratory screening on commodity hardware. By enabling the qualitative verification of synthetic coughs through time-domain reconstruction, this work also provides a pathway toward more interpretable and clinically trusted medical AI.

These results should be read in the context of the limitations stated in Section 7. The present study is a proof of concept validated at a single clinical site, and the full cascade has not yet been evaluated in a prospective multi-centre clinical cohort. The retrospective end-to-end analysis in Section 5.4 represents the closest such assessment possible without new prospective data collection.

Future work. Several directions remain for extending this research. Expanding the disease taxonomy to include conditions such as chronic obstructive pulmonary disease and tuberculosis would broaden clinical applicability. The one-vs-rest SVM architecture of the DAM is inherently extensible; new classes can be integrated by training additional binary classifiers without retraining existing modules. Future efforts will also focus on refining the generative fidelity of the VAE ensemble to capture more subtle acoustic biomarkers of disease progression. Furthermore, we intend to explore longitudinal monitoring and the integration of additional non-invasive modalities, such as lung auscultation and speech characteristics. Such a multimodal approach, combined with acoustically interpretable generative models, would substantially improve screening reliability and clinical utility in diverse healthcare settings.

CRedit authorship contribution statement

George Kontogiannis: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Pantelis Tzamalís:** Writing – review & editing. **Anastases Giannikopoulos:** Software. **Sotiris Nikolettseas:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] P. Kapetanidis, F. Kalioras, C. Tsakonas, P. Tzamalís, G. Kontogiannis, T. Karamanidou, T.G. Stavropoulos, S. Nikolettseas, Respiratory diseases diagnosis using audio analysis and artificial intelligence: a systematic review, *Sensors* 24 (4) (2024) 1173.
- [2] J. Korpáš, J. Sadloňová, M. Vrabec, Analysis of the cough sound: an overview, *Pulm. pharmacol.* 9 (5–6) (1996) 261–268.
- [3] P. Honkoop, O. Usmani, M. Bonini, The current and future role of technology in respiratory care, *Pulm. Ther.* 8 (2) (2022) 167–179.
- [4] F. Barata, D. Cleres, P. Tinschert, C.-H.I. Shih, F. Rassouli, M. Boesch, M. Brutsche, E. Fleisch, et al., Nighttime continuous contactless smartphone-based cough monitoring for the ward: validation study, *JMIR Form. Res.* 7 (1) (2023) e38439.
- [5] M. Galvosas, J.C. Gabaldón-Figueira, E.M. Keen, V. Orrillo, I. Blavia, J. Chaccour, P.M. Small, G. Giménez, M. Rudd, S.G. Lapierre, et al., Performance evaluation of the smartphone-based AI cough monitoring app-hyfe cough tracker against solicited respiratory sounds, *F1000research* 11 (2023) 730.
- [6] S. Valentine, B. Klasmer, M. Dabbah, M. Balabanovic, D. Plans, Smartphone Movement Sensors for Remote Monitoring of Respiratory Rate: Observational Study, *medRxiv*, pages 2021–12, 2021.

- [7] M. Faezipour, M. Faezipour, Sustainable smartphone-based healthcare systems: a systems engineering approach to assess the efficacy of respiratory monitoring apps, *Sustainability* 12 (12) (2020) 5061.
- [8] J.C. Galdon-Figueira, J. Brew, D.H. Doré, N. Umashankar, J. Chaccour, V. Orrillo, L.Y. Tsang, I. Blavia, A. Fernández-Montero, J. Bartolomé, et al., Digital acoustic surveillance for early detection of respiratory disease outbreaks in Spain: a protocol for an observational study, *BMJ Open* 11 (7) (2021) e051278.
- [9] M. Eni, V. Mordoh, Y. Zigel, Cough detection using a non-contact microphone: a nocturnal cough study, *PLOS ONE* 17 (1) (2022) e0262240.
- [10] S.S. Birring, T. Fleming, S. Matos, A.A. Raj, D.H. Evans, I.D. Pavord, The Leicester cough monitor: preliminary validation of an automated cough detection system in chronic cough, *Eur. Respir. J.* 31 (5) (2008) 1013–1018.
- [11] K. McGuinness, K. Holt, R. Dockry, J. Smith, P159 validation of the VitaloJAK™24 h ambulatory cough monitor, *Thorax* 67 (Suppl 2) (2012) A131.
- [12] J.I. Hall, M. Lozano, L. Estrada-Petrocelli, S. Birring, R. Turner, The present and future of cough counting tools, *J. Thorac. Dis.* 12 (9) (2020) 5207.
- [13] M. You, W. Wang, Y. Li, J. Liu, X. Xu, Z. Qiu, Automatic cough detection from realistic audio recordings using C-BiLSTM with boundary regression, *Biomed. Signal Process. Control* 72 (2022) 103304.
- [14] B.T. Atmaja, S. Zanjabila, A. Sasou, Comparing hysteresis comparator and RMS threshold methods for automatic single cough segmentations, *Int. J. Inf. Technol.* 16 (1) (2024) 5–12.
- [15] Y. Wang, M. Wahab, T. Hong, K. Molinari, G.M. Gauvreau, R.P. Cusack, Z. Gao, I. Satia, Q. Fang, Automated cough analysis with convolutional recurrent neural network, *Bioengineering* 11 (11) (2024) 1105.
- [16] K. Tan, J. Smith, P. Gaydecki, Cough sound spectro-temporal analysis and automated detection using vision transformers, *Digit. Health* 11 (2025) 20552076251394623, <https://doi.org/10.1177/20552076251394623>
- [17] J. Monge-Álvarez, C. Hoyos-Barceló, L.M. San-José-Reuelta, P. Casaseca-de-la Higuera, A machine hearing system for robust cough detection based on a high-level representation of band-specific audio features, *IEEE Trans. Biomed. Eng.* 66 (8) (2018) 2319–2330.
- [18] L. Orlando, T. Teijeiro, D. Atienza, The COUGHVID crowdsourcing dataset, a corpus for the study of large-scale cough analysis algorithms, *Sci. Data* 8 (1) (2021) 156.
- [19] T. Xia, D. Spathis, J. Ch, A. Grammenos, J. Han, A. Hasthanasombat, E. Bondareva, T. Dang, A. Floto, P. Cicuta, et al., COVID-19 sounds: a large-scale audio dataset for digital respiratory screening, in: *Thirty-Fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*, 2021.
- [20] G. Chaudhari, X. Jiang, A. Fakhry, A. Han, J. Xiao, S. Shen, A. Khanzada, Virufy: global applicability of crowdsourced and clinical datasets for AI detection of COVID-19 from cough, *arXiv preprint arXiv:2011.13320*, 2020.
- [21] E. Bair, Semi-supervised clustering methods, *Wiley Interdiscip. Rev. Comput. Stat.* 5 (5) (2013) 349–361.
- [22] K.J. Piczak, ESC: dataset for environmental sound classification, in: *Proceedings of the 23rd ACM International Conference on Multimedia*, 2015, pp. 1015–1018.
- [23] P. Tzamalís, A. Bardoutsos, D. Markantonatos, C. Raptopoulos, S. Nikoletsas, X. Aggelides, N. Papadopoulos, End-to-end gesture recognition framework for the identification of allergic rhinitis symptoms, in: *2022 18th International Conference on Distributed Computing in Sensor Systems (DCOSS)*, IEEE, 2022, pp. 25–34.
- [24] D.M.W. Powers, Evaluation: from precision, recall and F-measure to ROC, informedness, markedness and correlation, *J. Mach. Learn. Technol.* 2 (1) (2011). Re-published on arXiv, 2020., 37–63, <https://arxiv.org/abs/2010.16061>.
- [25] M. Kuluozturk, M.A. Kobat, P.D. Barua, S. Dogan, T. Tuncer, R.-S. Tan, E.J. Ciaccio, U.R. Acharya, DKPNet41: directed knight pattern network-based cough sound classification model for automatic disease diagnosis, *Med. Eng. Phys.* 110 (2022) 103870.
- [26] S. Hegde, S. Sreeram, I.L. Alter, C. Shor, T.A. Valdez, K.D. Meister, A. Rameau, Cough sounds in screening and diagnostics: a scoping review, *Laryngoscope* 134 (3) (2024) 1023–1031.
- [27] D. Bhattacharya, N.K. Sharma, D. Dutta, S.R. Chetupalli, P. Mote, S. Ganapathy, C. Chandrakiran, S. Nori, K.K. Suhail, S. Gonuguntla, et al., Coswara: a respiratory sounds and symptoms dataset for remote screening of SARS-CoV-2 infection, *Sci. Data* 10 (1) (2023) 397.
- [28] A. Pal, M. Sankarasubbu, Pay attention to the cough: early diagnosis of COVID-19 using interpretable symptoms embeddings with cough sound signal processing, in: *Proceedings of the 36th Annual ACM Symposium on Applied Computing*, 2021, pp. 620–628.
- [29] C. Brown, J. Chauhan, A. Grammenos, J. Han, A. Hasthanasombat, D. Spathis, T. Xia, P. Cicuta, C. Mascolo, Exploring automatic diagnosis of COVID-19 from crowdsourced respiratory sound data, in: *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2020, pp. 3474–3484.
- [30] P. Bagad, A. Dalmia, J. Doshi, A. Nagrani, P. Bhamare, A. Mahale, S. Rane, N. Agarwal, R. Panicker, Cough against covid: evidence of COVID-19 signature in cough sounds, *arXiv preprint arXiv:2009.08790*, 2020.
- [31] G. Celik, CovidCoughNet: a new method based on convolutional neural networks and deep feature extraction using pitch-shifting data augmentation for COVID-19 detection from cough, breath, and voice signals, *Comput. Biol. Med.* 163 (2023) 107153.
- [32] S. Han, Y.-R. Lee, J.-H. Lee, J. Jeon, C. Min, K. Kim, D. Kim, M.P. Kim, Y.M. Park, U. An, et al., Deep learning-based cough classification using application-recorded sounds: a transfer learning approach with VGGish, *BMC Med. Inform. Decis. Mak.* 25 (1) (2025) 228.
- [33] S.B. Sangle, P.H. Kachare, J. Nirmal, M. Alhameed, I. Al-Shourbarji, Warped-frequency cepstral coefficients for improving COVID-19 detection, in: *International Conference on the AI Revolution*, Springer, 2025, pp. 294–304.
- [34] G. Datkhile, P.H. Kachare, S.B. Sangle, I. Al-Shourbarji, A. Jabbari, R. Kirner, A. Alameen, Speech-based respiratory diagnostics: a study on COVID-19 detection with machine learning, *PLOS ONE* 20 (11) (2025) e0332146.
- [35] D.V. Puri, P.H. Kachare, I. Al-Shourbarji, A. Jabbari, R. Kirner, A. Alameen, Design of efficient bi-orthogonal wavelets for EEG-based detection of schizophrenia, *Eng. Sci. Technol. Int. J.* 68 (2025) 102090.
- [36] D.V. Puri, P.H. Kachare, S.B. Sangle, I. Al-Shourbarji, A. Jabbari, R. Kirner, A hybrid dwarf mongoose and artificial ecosystem-based EEG channel optimization for schizophrenia detection, *Biomed. Signal Process. Control* 112 (2026) 108513.
- [37] D. Puri, P. Kachare, S. Khare, I. Al-Shourbarji, A. Jabbari, A. Alameen, Hybrid reptile-snake optimizer based channel selection for enhancing Alzheimer's disease detection, *J. Bionic Eng.* 22 (2) (2025) 884–900.
- [38] S. Wei, S. Zou, F. Liao, et al., A comparison on data augmentation methods based on deep learning for audio classification, in: *Journal of Physics: Conference Series*, 1453, IOP Publishing, 2020, page 012085.
- [39] J. Salamon, J.P. Bello, Deep convolutional neural networks and data augmentation for environmental sound classification, *IEEE Signal Process. Lett.* 24 (3) (2017) 279–283.
- [40] D.S. Park, W. Chan, Y. Zhang, C.-C. Chiu, B. Zoph, E.D. Cubuk, Q.V. Le, SpecAugment: a simple data augmentation method for automatic speech recognition, *arXiv preprint arXiv:1904.08779*, 2019.
- [41] D. Naranjo, J. Chica, C. Salamea Palacios, Effects of data augmentation on the identification of cough sound using convolutional neural networks, in: A. Rocha, P.C. López-López, J.P. Salgado-Guerrero (Eds.), *Communication, Smart Technologies and Innovation for Society*, Springer Singapore, Singapore, 2022, pp. 285–295. ISBN 978-981-16-4126-8.
- [42] V. Ramesh, K. Vatanparvar, E. Nemati, V. Nathan, M.M. Rahman, J. Kuang, Coughgan: generating synthetic coughs that improve respiratory disease classification, in: *2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, IEEE, 2020, pp. 5682–5688.
- [43] N. Yella, B. Rajan, Data augmentation using GAN for sound based COVID 19 diagnosis, in: *2021 11th IEEE International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications (IDAACS)*, 2, IEEE, 2021, pp. 606–609.
- [44] C. Donahue, J. McAuley, M. Puckette, Adversarial audio synthesis, *arXiv preprint arXiv:1802.04208*, 2018.
- [45] S. Vasquez, M. Lewis, Melnet: a generative model for audio in the frequency domain, *arXiv preprint arXiv:1906.01083*, 2019.
- [46] K. Sohn, H. Lee, X. Yan, Learning structured output representation using deep conditional generative models, *Adv. Neural Inf. Process. Syst.* 28 (2015).
- [47] I. Higgins, L. Matthey, A. Pal, C.P. Burgess, X. Glorot, M.M. Botvinick, S. Mohamed, A. Lerchner, beta-vae: learning basic visual concepts with a constrained variational framework, *ICLR (Poster)* 3 (2017).
- [48] A. Van Den Oord, O. Vinyals, et al., Neural discrete representation learning, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [49] K. Kilgour, M. Zuluaga, D. Roblek, M. Sharifi, Fr'echet audio distance: A metric for evaluating music enhancement algorithms, *arXiv preprint arXiv:1812.08466*, 2018.
- [50] D.P. Kingma, Auto-encoding variational Bayes, *arXiv preprint arXiv:1312.6114*, 2013.
- [51] A. Van Den Oord, N. Kalchbrenner, K. Kavukcuoglu, Pixel recurrent neural networks, in: *International Conference on Machine Learning*, PMLR, 2016, pp. 1747–1756.
- [52] D. Griffin, J. Lim, Signal estimation from modified short-time Fourier transform, *IEEE Trans. Acoust. Speech Signal Process.* 32 (2) (1984) 236–243.
- [53] G. Kontogiannis, P. Tzamalís, S. Nikoletsas, Exploring the impact of synthetic data on human gesture recognition tasks using GANs, in: *2024 20th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT)*, IEEE, 2024, pp. 384–391.
- [54] J. Saldanha, S. Chakraborty, S. Patil, K. Kotecha, S. Kumar, A. Nayyar, Data augmentation using variational autoencoders for improvement of respiratory disease classification, *PLOS ONE* 17 (8) (2022) e0266467.
- [55] C. Esteban, S.L. Hyland, G. Rätsch, Real-valued (medical) time series generation with recurrent conditional GANs, *arXiv preprint arXiv:1706.02633*, 2017.
- [56] S. Dash, A. Yale, I. Guyon, K.P. Bennett, Medical time-series data generation using generative adversarial networks, in: *Artificial Intelligence in Medicine: 18th International Conference on Artificial Intelligence in Medicine, AIME 2020, Minneapolis, MN, USA, August 25–28, 2020, Proceedings 18*, Springer, 2020, pp. 382–391.
- [57] R. Kubichek, Mel-cepstral distance measure for objective speech quality assessment, in: *Proceedings of IEEE Pacific Rim Conference on Communications Computers and Signal Processing*, 1, IEEE, 1993, pp. 125–128.
- [58] P. Tzamalís, G. Kontogiannis, K. Giannopoulos, S. Nikoletsas, CrossAI time series: an open-source Python library for developing end-to-end AI pipelines for time series analysis, such as audio, motion, and other uni- or multi-axes tasks, <https://github.com/AIoT-Group-UoP/crossai-ts>.
- [59] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al., Scikit-Learn: machine learning in Python, *J. mach. Learn. res.* 12 (2011) 2825–2830.

- [60] M. Pahar, M. Klopfer, B. Reeve, R. Warren, G. Theron, A. Diacon, T. Niesler, Automatic tuberculosis and COVID-19 cough classification using deep learning, in: 2022 International Conference on Electrical, Computer and Energy Technologies (ICECET), IEEE, 2022, pp. 1–9.
- [61] A. Cattamanchi, S. Grandjean Lapierre, G. Theron, W. Worodria, A. Andama, C. Yu, N. Viet Nhung, D.J. Christopher, M. Raberahona, R. Rakotoarivelo, I. Lyimo, O. Lweno, CODA TB DREAM challenge dataset (synapse: syn31472953). Synapse Commons Platform,, 2024, <https://www.synapse.org/Synapse:syn31472953>.
- [62] M. Sharma, V. Nduba, L.N. Njagi, W. Murithi, Z. Mwongera, T.R. Hawn, S.N. Patel, D.J. Horne, TBscreen: a passive cough classifier for tuberculosis screening with a controlled dataset, Sci. Adv. 10 (1) (2024) eadi0282, <https://doi.org/10.1126/sciadv.adi0282>